READING 5

CURRENCY EXCHANGE RATES: UNDERSTANDING EQUILIBRIUM VALUE

EXAM FOCUS

There's no fluff here; you need it all. Take it slow and get a good understanding of quotes, currency cross rates, triangular arbitrage, all parity conditions, and their interrelationships. Forecasting exchange rates has important applications for valuation (which is the focus of Level II). Accordingly, theories of exchange rate determination as well as factors influencing exchange rates are all important. Be prepared to identify warning signs of currency crises.

MODULE 5.1: FOREX QUOTES, SPREADS, AND TRIANGULAR ARBITRAGE

Video covering this content is available online.

LOS 5.a: Calculate and interpret the bid–offer spread on a spot or forward currency quotation and describe the factors that affect the bid–offer spread.

PROFESSOR'S NOTE

The "bid–offer" spread is also known as the "bid–ask" spread: the terms "ask" and "offer" mean the same thing. Accordingly, we will be using them interchangeably.

Exchange Rates

An **exchange rate** is simply the price of one currency in terms of another. For example, a quote of 1.4126 USD/EUR means that each euro costs \$1.4126. In this example, the euro is called the *base* currency and the USD the *price* currency. Hence, a quote is the price of one unit of the base currency in terms of the price currency.

A **spot exchange rate** is the currency exchange rate for immediate delivery, which for most currencies means the exchange of currencies takes place two days after the trade. A **forward exchange rate** is a currency exchange rate for an exchange to be done in the future. Forward rates are quoted for various future dates (e.g., 30 days, 60 days, 90 days, or one year). A forward contract is an agreement to exchange a specific amount of one currency for a specific amount of another currency on a future date specified in the forward agreement.

Dealer quotes often include both bid and offer (ask) rates. For example, the euro could be quoted as \$1.4124 − 1.4128. The bid price (\$1.4124) is the price at which the dealer will buy euros, and the offer price (\$1.4128) is the price at which the dealer will sell euros.

Foreign Exchange Spread

The difference between the offer and bid price is called the *spread*. Spreads are often stated as "pips." When the spot quote has four decimal places, one pip is 1/10,000. In the previous example, the spread is \$0.0004 (4 pips) reflecting the dealer's profit. Dealers manage their foreign currency inventories by transacting in the interbank market (think of this as a wholesale market for currency). Spreads are narrow in the interbank market.

The spread quoted by a dealer depends on:

- **The spread in an interbank market for the same currency pair.** Dealer spreads vary directly with spreads quoted in the interbank market.
- **The size of the transaction.** Larger, liquidity-demanding transactions generally get quoted a larger spread.
- **The relationship between the dealer and client.** Sometimes dealers will give favorable rates to preferred clients based on other ongoing business relationships.

The interbank spread on a currency pair depends on:

- **Currencies involved.** Similar to stocks, high-volume currency pairs (e.g., USD/EUR, USD/JPY, and USD/GBP) command lower spreads than do lower-volume currency pairs (e.g., AUD/CAD).
- **Time of day.** The time overlap during the trading day when both the New York and London currency markets are open is considered the most liquid time window; spreads are narrower during this period than at other times of the day.
- **Market volatility.** Spreads are directly related to the exchange rate volatility of the currencies involved. Higher volatility leads to higher spreads to compensate market makers for the increased risk of holding those currencies. Spreads change over time in response to volatility changes.

In addition to these factors, spreads in forward exchange rate quotes increase with maturity. The reasons for this are: longer maturity contracts tend to be less liquid, counterparty credit risk in forward contracts increases with maturity, and interest rate risk in forward contracts increases with maturity.

Warm-Up: Working with Foreign Exchange Quotes

Earlier, we stated that a dealer will sell a currency at the ask price and purchase it at the bid price. We need to be a bit more specific. For example, imagine that you are given a USD/AUD bid and ask quote of 1.0508-1.0510. Investors can buy AUD (i.e., the base currency) from the dealer at the ask price of USD 1.0510. Similarly, investors can sell AUD to the dealer at the bid price of USD 1.0508. Remember, investors always take a loss due to the spread. So the rule is *buy the base currency at ask, and sell the base currency at bid*.

For transactions in the price currency, we do the opposite. If we need to buy USD (i.e., the price currency) using AUD (i.e., selling the base currency), we now use the dealer *bid* quote. Similarly, to sell the price currency, we use the dealer *ask* quote. So the rule is *buy the price currency at bid, and sell the price currency at ask*.

Alternatively, it is useful to follow the *up-the-bid-and-multiply*, *down-the-ask-and-divide rule*. Again given a USD/AUD quote, if you want to convert USD into AUD (you are going down the quote—from USD on top to AUD on bottom), use the *ask* price for that quote. Conversely, if you want to convert AUD into USD, you are going up the quote (from bottom to top) and, hence, use the *bid* price.

EXAMPLE: Converting currencies using spot rates

A dealer is quoting the AUD/GBP spot rate as 1.5060 − 1.5067. How would we:

1. Compute the proceeds of converting 1 million GBP.
2. Compute the proceeds of converting 1 million AUD.

Answer:

1. To convert 1 million GBP into AUD, we go "up the quote" (i.e., from GBP in the denominator to AUD in the numerator). Hence, we would use the *bid* price of 1.5060 and multiply.

   1 million GBP × 1.5060 = 1,506,000 AUD

2. To convert 1 million AUD into GBP, we go "down the quote" (i.e., from AUD in the numerator to GBP in the denominator). Hence, we would use the *ask* price of 1.5067 and divide.

   1 million AUD / 1.5067 = 663,702.13 GBP

LOS 5.b: Identify a triangular arbitrage opportunity and calculate its profit, given the bid–offer quotations for three currencies.

Cross Rate

The **cross rate** is the exchange rate between two currencies implied by their exchange rates with a common third currency. It is necessary to use cross rates when there is no active foreign exchange (FX) market in the currency pair being considered. The cross rate must be computed from the exchange rates between each of these two currencies and a major third currency, usually the USD or EUR.

Suppose we have the following quotes:

USD/AUD = 0.60 and MXN/USD = 10.70. What is the cross rate between Australian dollars and pesos (MXN/AUD)?

$$\frac{\text{MXN}}{\text{AUD}} = \frac{\cancel{\text{USD}}}{\text{AUD}} \times \frac{\text{MXN}}{\cancel{\text{USD}}} = 0.60 \times 10.70 = 6.42$$

So our MXN/AUD cross rate is 6.42 pesos per Australian dollar. The key to calculating cross rates is to make sure the common currency cancels out.

Cross Rates with Bid–Ask Spreads

Bid–ask spreads complicate the calculation of cross rates considerably. Suppose we are given three currencies A, B, and C; we can have three pairs of currencies (i.e., A/B, A/C, and B/C).

Rules:

$$\left(\frac{A}{C}\right)_{bid} = \left(\frac{A}{B}\right)_{bid} \times \left(\frac{B}{C}\right)_{bid}$$

$$\left(\frac{A}{C}\right)_{offer} = \left(\frac{A}{B}\right)_{offer} \times \left(\frac{B}{C}\right)_{offer}$$

To compute the cross rate for A/C, given A/B and B/C, we can follow these rules to obtain the bid and offer prices. If we are instead given A/B and C/B rates, we will have to make adjustments to obtain the B/C bid and offer rates from the C/B bid and offer rates, because A/B × C/B ≠ A/C. The process is as follows:

$$\left(\frac{B}{C}\right)_{bid} = \frac{1}{\left(\frac{C}{B}\right)_{offer}}$$

$$\left(\frac{B}{C}\right)_{offer} = \frac{1}{\left(\frac{C}{B}\right)_{bid}}$$

Triangular Arbitrage

Real-world currency dealers will maintain bid/ask quotes that ensure a profit to the dealer, regardless of which currencies customers choose to trade. If this was not the case, customers could earn profits through the process of triangular arbitrage. In **triangular arbitrage**, we begin with three pairs of currencies, each with bid and ask quotes, and construct a triangle where each node in the triangle represents one currency. To check for arbitrage opportunities, we go around the triangle clockwise (and later, counterclockwise) until we reach our starting point. As before, we follow the up-the-bid-and-multiply, down-the-ask-and-divide rule.

The following example will illustrate triangular arbitrage.

EXAMPLE: Triangular arbitrage

The following quotes are available from the interbank market:

Quotes:

USD/AUD 0.6000 − 0.6015

USD/MXN 0.0933 − 0.0935

1. Compute the implied MXN/AUD cross rate.
2. If your dealer quotes MXN/AUD = 6.3000 − 6.3025, is an arbitrage profit possible? If so, compute the arbitrage profit in USD if you start with USD 1 million.

Answer:

1. To compute implied cross rates, we need:

$$\left(\frac{MXN}{AUD}\right)_{bid} = \left(\frac{USD}{AUD}\right)_{bid} \times \left(\frac{MXN}{USD}\right)_{bid}$$

Since we are given USD/MXN quotes instead of MXN/USD quotes, we first invert these quotes:

$$\left(\frac{MXN}{USD}\right)_{bid} = \frac{1}{\left(\frac{USD}{MXN}\right)_{offer}} = \left(\frac{1}{0.0935}\right) = 10.70 \text{ MXN/USD}$$

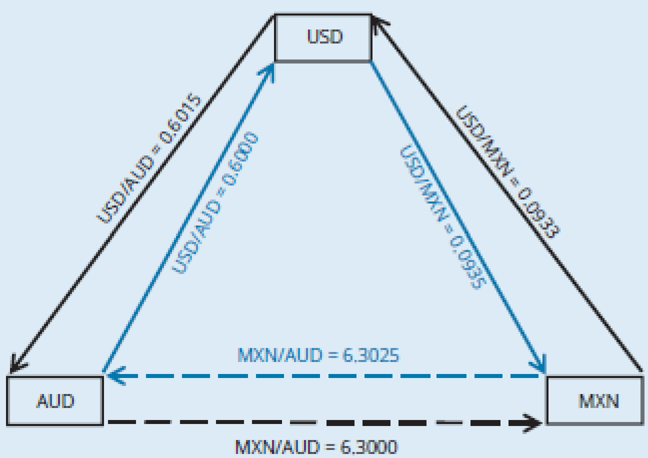
and

$$\left(\frac{MXN}{USD}\right)_{offer} = \frac{1}{\left(\frac{USD}{MXN}\right)_{bid}} = \left(\frac{1}{0.0933}\right) = 10.72 \text{ MXN/USD}$$

Now, the implied cross rates:

$$\left(\frac{MXN}{AUD}\right)_{bid} = \left(\frac{USD}{AUD}\right)_{bid} \times \left(\frac{MXN}{USD}\right)_{bid} = 0.60 \times 10.70 = 6.42$$
$$\left(\frac{MXN}{AUD}\right)_{offer} = \left(\frac{USD}{AUD}\right)_{offer} \times \left(\frac{MXN}{USD}\right)_{offer} = 0.6015 \times 10.72 = 6.4481$$

2. Since the dealer quote of MXN/AUD = 6.3000 − 6.3025 falls outside of these cross rates, arbitrage profit may be possible (we have to check this). Remember to use the dealer quotes in the triangle and not the cross rates we computed.



To label the arrows in this triangle, we follow the "up the bid, down the offer" rule. To convert from USD to MXN, ("down" with respect to the USD/MXN quote), we use the offer rate of 0.0935.

Going clockwise and starting with USD 1 million:

1. Convert USD 1 million into MXN @ 0.0935 USD/MXN. Note that the quote is USD/MXN and hence we are going down, and thus need to use the ask. Also remember: down, divide. We get 1 million/0.0935 = 10,695,187 MXN.
2. Next, we convert 10,695,187 MXN into AUD @ 6.3025 MXN/AUD to get 1,696,975 AUD.

3. Finally, we convert AUD 1,696,975 into USD @ 0.6000 USD/AUD. Here the quote is USD/AUD and we are converting from AUD to USD, so we are going "up the quote" and need to multiply by the bid. (Remember: up, multiply.) We get 1,696,975 × 0.60 = 1,018,185 USD − a profit of 18,185 USD.

We can also check for arbitrage in the counter-clockwise direction (even though we can never earn an arbitrage profit in both directions):

1. Convert USD 1 million into AUD using 0.6015. Again, the quote is USD/AUD and we are going down, so use the ask price and divide. We get 1 million/0.6015 = 1,662,510 AUD.
2. Next, we convert 1,662,510 AUD into MXN using 6.3000 to get 10,473,814 MXN.
3. Finally, we convert MXN 10,473,814 into USD at 0.0933 to get 977,207 USD − a loss of 22,793 USD.

LOS 5.c: Explain spot and forward rates and calculate the forward premium/discount for a given currency.

A currency is quoted at a **forward premium** relative to a second currency if the forward price (in units of the second currency) is *greater* than the spot price. A currency is quoted at a **forward discount** relative to a second currency if the forward price (in units of the second currency) is *less* than the spot price. The premium or discount is for the base currency (i.e., the currency at the bottom of the quote). For example, if the spot price is 1.20\$/€ and the forward price is 1.25\$/€, we say that the euro is trading at a forward premium.

$$\text{forward premium (discount)} = F - S_0$$

Given a quote of A/B, if this equation results in a positive value, we say that currency B (i.e., the base currency) is trading at a *premium* in the forward market.

In the FX markets, forward quotes are often presented as a premium or discount over spot rates. The following example illustrates this convention.

EXAMPLE: Spot and forward quotes

Given the following quotes for AUD/CAD, compute the bid and offer rates for a 30-day forward contract.

Maturity	Rate
Spot	1.0511/1.0519
30-day	+3.9/+4.1
90-day	+15.6/+16.8
180-day	+46.9/+52.3

Answer:

Since the forward quotes presented are all positive, the CAD (i.e., the base currency) is trading at a forward *premium*.

30-day bid = 1.0511 + 3.9/10,000 = 1.05149

30-day offer = 1.0519 + 4.1/10,000 = 1.05231

The 30-day all-in forward quote for AUD/CAD is 1.05149/1.05231.

PROFESSOR'S NOTE

For an investor wishing to convert AUD into CAD in the forward market, the relevant quote would be the ask rate (remember the "down-the-ask" rule) of 1.05231. This is also known as the all-in (i.e., after adding (subtracting) the forward premium (discount) rate for the investor in question.

MODULE QUIZ 5.1

1. All of the following factors are likely to contribute to an increase in USD/EUR dealer spread except:

   A. increase in the volatility of EUR/USD spot rate.
   B. increase in the EUR/USD spread in the interbank market.
   C. smaller order size.

2. The bid–ask quotes for the USD, GBP, and EUR are:

   EUR/USD: 0.7000 – 0.7010

   USD/GBP: 1.7000 – 1.7010

   EUR/GBP: 1.2000 – 1.2010

   The potential arbitrage profit from a triangular arbitrage based on an initial position of 1 million USD is *closest* to:

   A. USD0.
   B. USD7,212.
   C. USD6,372.

MODULE 5.2: MARK-TO-MARKET VALUE, AND PARITY CONDITIONS

Video covering this content is available online.

LOS 5.d: Calculate the mark-to-market value of a forward contract.

If the forward contract price is consistent with covered interest rate parity (discussed later), the value of the contract at initiation is zero to both parties. After initiation, the value of the forward contract will change as forward quotes for the currency pair change in the market.

Mark-to-Market Value

The value of a forward currency contract prior to expiration is also known as the *mark-to-market value*. To compute the value of a forward contract prior to expiration, we take the difference between the forward price we locked-in and the current forward price, multiply that by the size of the contract, and then discount for the time period remaining until the contract settlement date.

$$V_t = \frac{(FP_t - FP)(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]}$$

where:

V_t = value of the forward contract at time t (to the party buying the base currency), ($t < T$) denominated in price currency

FP_t = forward price (to sell base currency) at time t in the market for a new contract maturing at time T

FP = forward price specified in the contract at inception (to buy the base currency)

days = number of days remaining to maturity of the forward contract ($T - t$)

R = interest rate of price currency

PROFESSOR'S NOTE

The interest rate references used throughout this reading are the rates on bank deposits using the appropriate market reference rate (MRR), and based on the days/360 convention. MRR is the generic label used in the CFA curriculum to represent various LIBOR successors globally.

EXAMPLE: Valuing a forward contract prior to maturity

Yew Mun Yip has entered into a 90-day forward contract long CAD 1 million against AUD at a forward rate of 1.05358 AUD/CAD. Thirty days after initiation, the following AUD/CAD quotes are available:

Maturity	FX Rate
Spot	1.0612/1.0614
30-day	+4.9/+5.2
60-day	+8.6/+9.0
90-day	+14.6/+16.8
180-day	+42.3/+48.3

The following information is available (at t = 30) for AUD interest rates:

30-day rate: 1.12%

60-day rate: 1.16%

90-day rate: 1.20%

What is the mark-to-market value in AUD of Yip's forward contract?

Answer:

Yip's contract calls for long CAD (i.e., converting AUD to CAD). To value the contract, we would look to unwind the position. To unwind the position, Yip can take an offsetting position in a new forward contract with the same maturity. Hence, Yip would be selling CAD in exchange for AUD and, hence, going up the bid (i.e., use the bid price). Note that after 30 days, 60 more days remain in the original contract.

The forward bid price for a new contract expiring in T − t = 60 days is 1.0612 + 8.6/10,000 = 1.06206.

The interest rate to use for discounting the value is also the 60-day AUD interest rate of 1.16%:

$$V_t = \frac{(FP_t - FP)\,(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]} = \frac{(1.06206 - 1.05358)\,(1{,}000{,}000)}{\left[1 + 0.0116\left(\frac{60}{360}\right)\right]} = 8{,}463.64$$

Thirty days into the forward contract, Yip's position has gained (positive value) AUD 8,463.64. This is because Yip's position is long CAD, which has appreciated relative to AUD since inception of the contract. Yip can close out the contract on that day and receive AUD 8,463.64.

Note: Be sure to use the AUD (price currency) interest rate.

LOS 5.e: Explain international parity conditions (covered and uncovered interest rate parity, forward rate parity, purchasing power parity, and the international Fisher effect).

Covered Interest Rate Parity

The word *covered* in the context of covered interest parity means bound by arbitrage. **Covered interest rate parity** holds when any forward premium or discount exactly offsets differences in interest rates, so that an investor would earn the same return investing in either currency. If euro interest rates are higher than dollar interest rates, the forward discount on the euro relative to the dollar will just offset the higher euro interest rate.

Formally, covered interest rate parity requires that (given A/B quote structure):

$$F = \frac{\left[1 + R_A\left(\frac{\text{days}}{360}\right)\right]}{\left[1 + R_B\left(\frac{\text{days}}{360}\right)\right]} S_0$$

where:
F = forward rate (quoted as A/B)
S_0 = spot rate (quoted as A/B)
days = number of days in the underlying forward contract
R_A = interest rate for Currency A
R_B = interest rate for Currency B

PROFESSOR'S NOTE
For all parity relations, follow the numerator-denominator rule. If you are given a USD/EUR quote, the USD interest rate should be in the numerator and the EUR interest rate in the denominator of the parity equation.

Recall that:

$$\text{forward premium (discount)} = F - S_0 = \left[\frac{1 + R_A\left(\frac{\text{days}}{360}\right)}{1 + R_B\left(\frac{\text{days}}{360}\right)} - 1\right] S_0$$

or

$$\text{forward premium (discount)} = F - S_0 = S_0\left[\frac{\left(\frac{\text{days}}{360}\right)}{1 + R_B\left(\frac{\text{days}}{360}\right)}\right](R_A - R_B)$$

EXAMPLE: Covered interest arbitrage

The U.S. dollar MRR is 8%, and the euro MRR is 6%. The spot exchange rate is $1.30 per euro (USD/EUR), and the 1-year forward rate is $1.35 per euro. Determine whether a profitable arbitrage opportunity exists, and illustrate such an arbitrage if it does.

Answer:

First, we note that the forward value of the euro is too high. Interest rate parity would require a forward rate of:

$$\$1.30(1.08 / 1.06) = \$1.3245$$

Because the market forward rate of $1.35 is higher than that implied by interest rate parity, we should sell euros in the forward market and do the opposite (i.e., buy euros) in the spot market. The steps in the covered interest arbitrage are as follows.

Initially:

Step 1: Borrow $1,000 at 8%.

Step 2: Purchase 1,000 / 1.30 = 769.23 euros in the spot market.

Step 3: Invest the euros at 6%.

Step 4: Enter into a forward contract to sell the expected proceeds at the end of one year (i.e., 769.23 × 1.06 = 815.38 euros), at $1.35 each.

After one year:

Step 1: Sell the 815.38 euros under the terms of the forward contract at $1.35 to get $1,100.76.

Step 2: Repay the $1,000 8% loan, which requires $1,080.

Step 3: Keep the difference of $20.76 as an arbitrage profit.

Uncovered Interest Rate Parity

With covered interest rate parity, arbitrage will force the forward contract exchange rate to a level consistent with the difference between the two country’s nominal interest rates. If forward currency contracts are not available, or if capital flows are restricted so as to prevent arbitrage, the relationship need not hold. **Uncovered interest rate parity** refers to such a situation; uncovered in this context means not bound by arbitrage.

Consider Country A where the interest rate is 4%, and Country B where the interest rate is 9%. Under uncovered interest rate parity, currency B is expected to depreciate by 5%

annually relative to currency A, so that an investor should be indifferent between investing in Country A or B.

Given a quote structure of A/B, the base currency (i.e., currency B) is expected to appreciate by approximately $R_A - R_B$. (When $R_A - R_B$ is negative, currency B is expected to depreciate). Mathematically:

$$E(\%\Delta S)_{(A/B)} = R_A - R_B$$

The following example illustrates the use of uncovered interest rate parity to *forecast* future spot exchange rates using market interest rates.

EXAMPLE: Forecasting spot rates with uncovered interest rate parity

Suppose the spot exchange rate quote is ZAR/EUR = 8.385. The 1-year nominal rate in the eurozone is 10% and the 1-year nominal rate in South Africa is 8%. Calculate the expected percentage change in the exchange rate over the coming year using uncovered interest rate parity.

Answer:

The rand interest rate is less than the euro interest rate, so uncovered interest rate parity predicts that the value of the rand will rise (it will take fewer rand to buy one euro) because of higher interest rates in the eurozone. The euro (the base currency) is expected to “appreciate” by approximately $R_{ZAR} - R_{EUR}$ = 8% − 10% = –2%. (Note the *negative* 2% value.) Thus the euro is expected to *depreciate* by 2% relative to the rand, leading to a change in exchange rate from 8.385 ZAR/EUR to 8.217 ZAR/EUR over the coming year.

Comparing covered and uncovered interest parity, we see that covered interest rate parity derives the *no-arbitrage forward rate*, while uncovered interest rate parity derives the *expected future spot rate* (which is not market traded). Covered interest parity is assumed by arbitrage, but this is not the case for uncovered interest rate parity.

Under uncovered interest rate parity, if the foreign interest rate is higher by 2%, the foreign currency is expected to depreciate by 2%, so the investor should be indifferent between investing in the foreign currency or in their own domestic currency. An investor that chooses to invest in the foreign currency without any additional return (the interest rate differential is offset by currency value changes) is not demanding a risk premium for the foreign currency risk. Hence, uncovered interest rate parity assumes that the investor is *risk-neutral*.

If the forward rate is equal to the expected future spot rate, we say that the forward rate is an **unbiased predictor** of the future spot rate. In such an instance, $F = E(S_1)$; this is called **forward rate parity**. In this special case, if covered interest parity holds (and it will; by arbitrage) uncovered interest parity would also hold (and vice versa). Stated differently, if uncovered interest rate parity holds, forward rate parity also holds (i.e., the forward rate is an unbiased predictor of the future spot rate).

There is no reason that uncovered interest rate parity must hold in the short run, and indeed it typically does not. There is evidence that it does generally hold in the long run, so longer-term expected future spot rates based on uncovered interest rate parity are often used as forecasts of future exchange rates.

(Domestic) Fisher Relation

Professor Irving Fisher originated the idea that the nominal rate of return is (approximately) the sum of the real rate and the expected rate of inflation.

We can write this relation (known as the Fisher relation) as:

$$R_{\text{nominal}} = R_{\text{real}} + E(\text{inflation})$$

International Fisher Relation

Under **real interest rate parity**, real interest rates are assumed to converge across different markets. Taking the Fisher relation and real interest rate parity together gives us the **international Fisher effect**:

$$R_{\text{nominal A}} - R_{\text{nominal B}} = E(\text{inflation}_A) - E(\text{inflation}_B)$$

This tells us that the difference between two countries' nominal interest rates should be equal to the difference between their expected inflation rates.

The argument for the equality of real interest rates across countries is based on the idea that with free capital flows, funds will move to the country with a higher real rate until real rates are equalized.

EXAMPLE: **Calculating the real interest rate**

Suppose the nominal South African interest rate is 9.0% and the expected inflation rate is 3.5%. Calculate the real interest rate.

Answer:

0.090 = real r_{ZAR} + 0.035

real r_{ZAR} = 0.090 − 0.035 = 0.055, or 5.5%

If we move to a 2-country scenario, we will now have two nominal interest rates and two expected inflation rates. If the real rates for both countries are assumed to be equal, they drop out of the equation, and we are left with the international Fisher relation, as shown in the following example.

EXAMPLE: **Using the international Fisher relation**

Suppose that the eurozone expected annual inflation rate is 9.0%, and that the expected South African inflation rate is 13.0%. The nominal interest rate is 10.09% in the eurozone. Use the international Fisher relation to estimate the nominal interest rate in South Africa.

Answer:

real rate ZAR = real rate EUR ≈ (nominal interest rate in the eurozone) – (eurozone expected annual inflation rate) = 10.09% − 9% = 1.09%

R_{ZAR} = (expected South African inflation rate) + (real ZAR interest rate)

= 13% + 1.09% = 14.09%

Purchasing Power Parity

The law of one price states that identical goods should have the same price in all locations. For instance, a pair of designer jeans should cost the same in Paris as they do in New York, after adjusting for the exchange rate. The potential for arbitrage is the basis for the law of one price: if designer jeans cost less in New York than they do in Paris, an enterprising individual will buy designer jeans in New York and sell them in Paris, until this action causes the price differential to disappear. Note, however, that the law of one price does not hold in practice, due to the effects of frictions such as tariffs and transportation costs.

Instead of focusing on individual products, **absolute purchasing power parity** (absolute PPP) compares the average price of a representative basket of consumption goods between countries using an index such as The United States **Consumer Price Index (CPI)**. Absolute PPP requires only that the law of one price be correct *on average*, that is, for like baskets of goods in each country.

S(A/B) = CPI(A) / CPI(B)

In practice, even if the law of one price held for every good in two economies, absolute PPP might not hold because the weights (consumption patterns) of the various goods in the two economies may not be the same (e.g., people eat more potatoes in Russia and more rice in Japan).

Relative Purchasing Power Parity

Relative purchasing power parity (relative PPP) states that changes in exchange rates should exactly offset the price effects of any inflation differential between two countries. Simply put, if (over a 1-year period) Country A has a 6% inflation rate and Country B has a 4% inflation rate, then Country A's currency should *depreciate* by approximately 2% relative to Country B's currency over the period.

The equation for relative PPP is as follows:

$$\%\Delta S_{(A/B)} = \text{Inflation}_{(A)} - \text{Inflation}_{(B)}$$

where:

$\%\Delta S_{(A/B)}$ = change in spot price (A/B)

Relative PPP is based on the idea that even if absolute PPP does not hold, there may still be a relationship between changes in the exchange rate and differences between the inflation rates of the two countries.

Ex-Ante Version of PPP

The ex-ante version of purchasing power parity is the same as relative purchasing power parity except that it uses *expected* inflation instead of actual inflation.

The following example illustrates the use of the ex-ante version of the PPP relation.

EXAMPLE: Calculating the exchange rate predicted by the ex-ante version of PPP

The current spot rate is USD/AUD = 1.00. You expect the annualized Australian inflation rate to be 5%, and the annualized U.S. inflation rate to be 2%. According to the ex-ante version of PPP, what is the expected change in the spot rate over the coming year?

Answer:

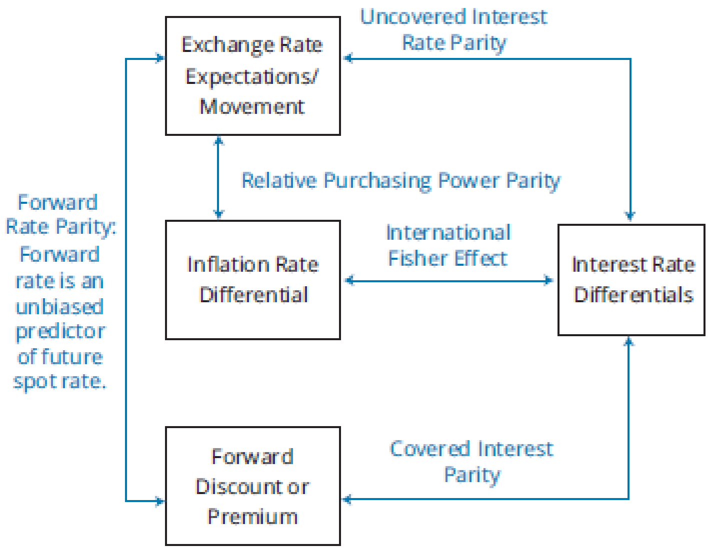
Since the AUD has the higher expected inflation rate, we expect that the AUD will depreciate relative to the USD. To keep the cost of goods and services the same across borders, countries with higher rates of inflation should see their currencies depreciate. The expected change in the spot rate over the coming year is inflation(USD) − inflation(AUD) = 2% − 5% = –3%. This predicts a new USD/AUD exchange rate of approximately 0.97 USD/AUD.

Because there is no true arbitrage available to force relative PPP to hold, violations of relative PPP in the short run are common. However, because the evidence suggests that the relative form of PPP holds approximately in the long run, it remains a useful method for estimating the relationship between exchange rates and inflation rates.

LOS 5.f: Describe relations among the international parity conditions.

It is useful to establish how all the parity relations described earlier fit together. Figure 5.1 shows the interrelationships among parity conditions. Though these relationships are not all exact, together they provide an extremely useful framework for thinking about exchange rates.

Figure 5.1: The International Parity Relationships Combined



Several observations can be made from the relationships among the various parity conditions:

- Covered interest parity holds by arbitrage. If forward rate parity holds, uncovered interest rate parity also holds (and vice versa).
- Interest rate differentials should mirror inflation differentials. This holds true if the international Fisher relation holds. If that is true, we can also use inflation differentials to forecast future exchange rates—which is the premise of the ex-ante version of PPP.
- If the ex-ante version of relative PPP as well as the international Fisher relation both hold, uncovered interest rate parity will also hold.

LOS 5.g: Evaluate the use of the current spot rate, the forward rate, purchasing power parity, and uncovered interest parity to forecast future spot exchange rates.

LOS 5.h: Explain approaches to assessing the long-run fair value of an exchange rate.

We can use ex-ante PPP, uncovered interest rate parity, or forward rates to forecast future spot rates. As stated earlier, uncovered interest rate parity and PPP are not bound by arbitrage and seldom work over the short and medium terms. Similarly, the forward rate is not an unbiased predictor of future spot rate. However, PPP holds over reasonably long time horizons. If relative PPP holds at any point in time, the real exchange rate (i.e., the exchange rate adjusted for relative historical inflation between the currency pair) would be constant. However, since relative PPP seldom holds over the short term, the real exchange rate fluctuates around its mean-reverting equilibrium value.

The international Fisher effect (and real rate parity) assumes that there are no differences between sovereign risk premia (i.e., all countries are perceived to be equally risky by investors). This is obviously untrue as investors do demand a higher real rate of return (i.e., a risk premium) for investing in emerging market currencies that are perceived to be riskier.

MODULE QUIZ 5.2

1. Suppose the spot exchange rate quote is 1.0120 Canadian dollars (C$) per U.S. dollar. The 1-year nominal interest rate in Canada is 3.0% and the 1-year nominal interest rate in the United States is 1.0%. The expected exchange rate at the end of the year using the uncovered interest rate parity is *closest* to:
   A. C$1.0322.
   B. C$0.9923.
   C. C$0.9918.
2. The international parity relationships indicate that the expected return on risk-free securities should be the same in all countries and exchange rate risk is really just inflation risk. Which of the following is *least likely* to be considered a practical implication of this framework?
   A. Investors will earn the same real rate of return on investments once their own currency impact is accounted for.
   B. Interest rate differentials reflect currency expectations. As a result, covered interest arbitrage will provide a return in any foreign currency that is equal to the domestic return.
   C. There are significant rewards for bearing foreign exchange risk.
3. For uncovered interest rate parity to hold, which condition is necessary?
   A. Forward rate parity holds.
   B. Covered interest rate parity holds and ex-ante relative PPP holds.
   C. Real interest rate parity and ex-ante relative PPP holds.

Use the following information to answer Questions 4 through 9.

Sally Franklin, CFA, is a financial advisor to Jamie Curtess, a U.S. citizen interested in learning more about how her investments will be affected by exchange rates and differences in interest rates internationally. Franklin has gathered the following information based on Curtess's investment interests.

The current spot exchange rate: $1 = €0.74.

	Europe	United States
Nominal 1-year interest rate:	4%	?
Expected annual inflation:	2%	1%

Franklin also gathers the following information:

	Switzerland	South Africa
Nominal 1-year interest rate:	5%	7%
Expected annual inflation:	3%	5%

4. According to the international Fisher relation, the 1-year nominal interest rate in the United States should be *closest* to:
   A. 3.00%.
   B. 4.34%.
   C. 6.00%.
5. If the relative form of the PPP holds, the expected exchange rate in one year is *closest* to:
   A. $1.3378 per €.
   B. $0.7463 per €.
   C. $1.3647 per €.
6. For this question only, assume that the U.S. interest rate is 3.5%. The 1-year forward rate should be *closest* to:
   A. $1.3647 per €.
   B. $0.7463 per €.

C. $1.3449 per €.

7. Curtess wonders how spot rates are expected to change in the future and asks the following question: "What are the implications for the South African rand relative to the Swiss franc under uncovered interest rate parity, and the implications for the euro relative to the U.S. dollar under the relative form of purchasing power parity?" Franklin responds by making two statements:

   Statement 1: The South African rand is expected to depreciate relative to the Swiss franc.

   Statement 2: The euro is expected to depreciate relative to the U.S. dollar.

   Based upon the underlying parity relationships cited, are Franklin's statements accurate?

   A. No, both statements are inaccurate.
   B. Yes, both statements are accurate.
   C. One statement is accurate and one is inaccurate.

8. For this question only, imagine that the nominal interest rate in the United States is 3%. Real interest rates, using the Fisher relation, are *most likely* to be:
   A. greater in the United States than in Europe.
   B. lower in Europe than in South Africa.
   C. equal among Europe, South Africa, Switzerland, and the United States.

9. A forecasted $/€ exchange rate in one year equal to the current 1-year forward rate is *most likely* to be based on the assumption that:
   A. absolute PPP holds.
   B. investors are risk neutral.
   C. real interest rate parity holds.

MODULE 5.3: EXCHANGE RATE DETERMINANTS, CARRY TRADE, AND CENTRAL BANK INFLUENCE

Video covering this content is available online.

LOS 5.i: Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade.

FX Carry Trade

Uncovered interest rate parity states that a currency with a high interest rate should depreciate relative to a currency with a lower interest rate, so that an investor would earn the same return investing in either currency. For example, suppose that short-term interest rates are 3% in the U.K. and 1% in the United States. Uncovered interest rate parity implies that the GBP should depreciate by 2% relative to the USD over the coming year.

However, uncovered interest rate parity is not bound by arbitrage. If the GBP depreciates by less than 2% (or even appreciates), an investor who has invested in the higher yielding GBP using funds borrowed in USD will earn excess profits. In a **FX carry trade**, an investor invests in a higher yielding currency using funds borrowed in a lower yielding currency. The lower yielding currency is called the *funding currency*.

Consider the following example.

EXAMPLE: Carry trade

Interest Rates	Currency Pair	Exchange Rates	
		Today	*One year later*
U.K. 3%	USD/GBP	1.50	1.50
U.S. 1%			

Compute the profit to an investor borrowing in the United States and investing in the U.K.

Answer:

$$\begin{aligned}\text{return} &= \text{interest earned on investment} - \text{funding cost} \\ &\quad - \text{currency depreciation} \\ &= 3\% - 1\% - 0\% \\ &= 2\%\end{aligned}$$

The FX carry trade attempts to capture an interest rate differential and is a bet *against* uncovered interest rate parity. Carry trades typically perform well during low-volatility periods. Sometimes, higher yields attract larger capital flows, which in turn lead to an economic boom and appreciation (instead of depreciation) of the higher yielding currency. This could make the carry trade even more profitable because the investor earns a return from currency appreciation in addition to the return from the interest rate spread.

Risks of the Carry Trade

As discussed earlier, the carry trade is profitable only if uncovered interest rate parity does not hold over the investment horizon. The risk is that the funding currency may appreciate significantly against the currency of the investment, which would reduce a trader's profit—or even lead to a loss. Furthermore, the return distribution of the carry trade is not normal; it is characterized by negative **skewness** and excess **kurtosis** (i.e., fat tails), meaning that the probability of a large loss is higher than the probability implied under a normal distribution. We call this high probability of a large loss the **crash risk** of the carry trade.

Crash risk stems from the carry trade's leveraged nature: an investor borrows a low-yielding (funding) currency and then invests in a high-yielding currency. As more investors follow and adopt the same strategy, the demand for high-yielding currency actually pushes its value *up*. However, with this herding behavior comes the risk that all investors may attempt to exit the trade at the same time. (This is especially true if investors use stop-loss orders in their carry trades.) During turbulent times, as investors exit their positions (i.e., a flight to safety), the high-yielding currency can experience a steep decline in value, generating large losses for traders pursuing FX carry trades.

LOS 5.j: Explain how flows in the balance of payment accounts affect currency exchange rates.

Balance of Payments

Balance-of-payments (BOP) accounting is a method used to keep track of transactions between a country and its international trading partners. It includes government transactions, consumer transactions, and business transactions. The BOP accounts reflect all payments and liabilities *to* foreigners as well as all payments and obligations received *from* foreigners.

The **current account** measures the exchange of goods, the exchange of services, the exchange of investment income, and unilateral transfers (gifts to and from other nations). The current account balance summarizes whether we are selling more goods and services to the rest of the world than we are buying from them (a current account surplus) or buying more from the rest of the world than we are selling to them (a current account deficit).

The **financial account** (also known as the **capital account**) measures the flow of funds for debt and equity investment into and out of the country.

When a country experiences a current account deficit, it must generate a surplus in its capital account (or see its currency depreciate). Capital flows tend to be the dominant factor influencing exchange rates in the short term, as capital flows tend to be larger and more rapidly changing than goods flows.

Influence of BOP on Exchange Rates

Current Account Influences

Current account deficits lead to a depreciation of domestic currency via a variety of mechanisms:

- **Flow supply/demand mechanism.** Current account deficits in a country increase the supply of that currency in the markets (as exporters to that country convert their revenues into their own local currency). This puts downward pressure on the exchange value of that currency. The decrease in the value of the currency *may* restore the current account deficit to a balance—depending on the following factors:
  - *The initial deficit*. The larger the initial deficit, the larger the depreciation of domestic currency needed to restore current account balance.
  - *The influence of exchange rates on domestic import and export prices*. As a country's currency depreciates, the cost of imported goods increases. However, some of the increase in cost may not be passed on to consumers.
  - *Price elasticity of demand of the traded goods*. If the most important imports are relatively price inelastic, the quantity imported will not change.
- **Portfolio balance mechanism.** Countries with current account surpluses usually have capital account deficits, which typically take the form of investments in countries with current account deficits. As a result of these flows of capital, investor countries may find

their portfolios' composition being dominated by few investee currencies. When investor countries decide to rebalance their investment portfolios, it can have a significant negative impact on the value of those investee country currencies.

- **Debt sustainability mechanism.** A country running a current account deficit may be running a capital account surplus by borrowing from abroad. When the level of debt gets too high relative to GDP, investors may question the sustainability of this level of debt, leading to a rapid depreciation of the borrower's currency.

Capital Account Influences

Capital account flows are one of the major determinants of exchange rates. As capital flows into a country, demand for that country's currency increases, resulting in appreciation. Differences in real rates of return tend to be a major determinant of the flow of capital: higher relative real rates of return attract foreign capital. Capital flows into a country may be needed to overcome a shortage of internal savings to fund investments needed for economic growth. However, capital flows in excess of needed investment capital pose several problems. This is especially problematic for emerging markets.

Excessive capital inflows into emerging markets create problems for those countries such as:

- Excessive real appreciation of the domestic currency.
- Financial asset and/or real estate bubbles.
- Increases in external debt by businesses or government.
- Excessive consumption in the domestic market fueled by credit.

Emerging market governments often counteract excessive capital inflows by imposing capital controls or by direct intervention in the foreign exchange markets. We will discuss this further in a subsequent LOS.

LOS 5.k: Explain the potential effects of monetary and fiscal policy on exchange rates.

Mundell-Fleming Model

Developed in early 1960s, the **Mundell-Fleming model** evaluates the short-term impact of monetary and fiscal policies on interest rates—and consequently on exchange rates. The model assumes that there is sufficient slack in the economy to handle changes in aggregate demand, and that inflation is not a concern. Accordingly, changes in inflation rates due to changes in monetary or fiscal policy are not explicitly modeled by the Mundell-Fleming model.

We will look at the implications of this model for flexible exchange rate regimes as well as for fixed exchange rate regimes.

Flexible Exchange Rate Regimes

In a flexible ("floating") exchange rate system, rates are determined by supply and demand in the foreign exchange markets. We will examine the influence of monetary and fiscal policies when international capital flows are relatively unrestricted (high mobility of capital) versus when capital flows are relatively restricted (low mobility of capital), both under a flexible exchange rate system.

High Capital Mobility

Expansionary **monetary policy** and expansionary **fiscal policy** are likely to have opposite effects on exchange rates. Expansionary monetary policy will reduce the real interest rate and, consequently, reduce the inflow of capital investment in physical and financial assets. This decrease in financial inflows (deterioration of the financial account) reduces the demand for the domestic currency, resulting in depreciation of the domestic currency. Restrictive monetary policy should have the opposite effect, increasing real interest rates and leading to an appreciation in the value of the domestic currency.

Expansionary fiscal policy (an increased deficit from lower taxes or higher government spending) will increase government borrowing and, consequently, real interest rates. An increase in real interest rates will attract foreign investment, improve the financial account, and consequently, *increase* the demand for the domestic currency.

Low Capital Mobility

Our discussion so far has assumed free flow of capital, which is a valid assumption with respect to developed markets. In emerging markets, however, capital flows may be restricted. In that case, the impact of trade imbalance on exchange rates (goods flow effect) is greater than the impact of interest rates (financial flows effect). In such a case, expansionary fiscal or monetary policy leads to increases in net imports, leading to depreciation of the domestic currency. Similarly, restrictive monetary or fiscal policy leads to an appreciation of domestic currency. Figure 5.2 summarizes the influence of fiscal and monetary policy on the value of the domestic currency.

Figure 5.2: Monetary and Fiscal Policy vs. the Value of the Domestic Currency

Monetary Policy/Fiscal Policy	Capital Mobility	
	High	Low
Expansionary/Expansionary	Uncertain	Depreciation
Expansionary/Restrictive	Depreciation	Uncertain
Restrictive/Expansionary	Appreciation	Uncertain
Restrictive/Restrictive	Uncertain	Appreciation

PROFESSOR'S NOTE

Candidates are often confused by the implication under the Mundell-Fleming model that a higher-interest-rate currency will appreciate relative to a lower-interest-rate currency, because this is exactly the opposite of what we learned under uncovered interest rate parity. Note, though, that uncovered interest rate parity assumed that real interest rates are equal globally, and thus that nominal interest rates merely mirror expected inflation. That condition no longer holds under the Mundell-Fleming model, which does not consider inflation.

Fixed Exchange Rate Regimes

Under a fixed exchange rate regime, the government fixes the rate of exchange of its currency relative to one of the major currencies.

An expansionary (restrictive) monetary policy would lead to depreciation (appreciation) of the domestic currency as stated previously. Under a fixed rate regime, the government would then have to purchase (sell) its own currency in the foreign exchange market. This action essentially reverses the expansionary (restrictive) stance.

This explains why, in a world with mobility of capital, governments cannot both manage exchange rates as well as pursue independent monetary policy. If the government wants to manage monetary policy, it must either let exchange rates float freely or restrict capital movements to keep them stable.

Monetary Approach to Exchange Rate Determination

Monetary models only take into account the effect of monetary policy on exchange rates (fiscal policy effects are not considered). With the Mundell-Fleming model, we assume that inflation (price levels) plays no role in exchange rate determination. Under monetary models, we assume that output is fixed, so that monetary policy primarily affects inflation, which in turn affects exchange rates. There are two main approaches to monetary models:

1. **Pure monetary model.** Under a pure monetary model, the PPP holds at any point in time and output is held constant. An expansionary (restrictive) monetary policy leads to an increase (decrease) in prices and a decrease (increase) in the value of the domestic currency. Therefore an x% increase in the money supply leads to an x% increase in price levels and then to an x% depreciation of domestic currency. The pure monetary approach does not take into account expectations about future monetary expansion or contraction.
2. **Dornbusch overshooting model.** This model assumes that prices are sticky (inflexible) in the short term and, hence, do not immediately reflect changes in monetary policy (in other words, PPP does not hold in the short term). The model concludes that exchange rates will overshoot the long-run PPP value in the short term. In the case of an expansionary monetary policy, prices increase, but over time. Expansionary monetary policy leads to a *decrease* in interest rates—and a larger-than-PPP-implied depreciation of the domestic currency due to capital outflows. In the long term, exchange rates gradually increase toward their PPP implied values.

Similarly, a restrictive monetary policy leads to excessive appreciation of the domestic currency in the short term, and then a slow depreciation toward the long-term PPP value.

Portfolio Balance Approach to Exchange Rate Determination

The portfolio balance approach focuses only on the effects of *fiscal* policy (and not monetary policy). While the Mundell-Fleming model focuses on the short-term implications of fiscal policy, the **portfolio balance approach** takes a long-term view and evaluates the effects of a sustained fiscal deficit or surplus on currency values.

When the government runs a fiscal deficit, it borrows money from investors. Under the portfolio balance approach, investors evaluate the debt based on expected risk and return. A sovereign debt investor would earn a return based on both the debt's yield and its currency return. (When we invest in a foreign-currency-denominated security, our realized return will be comprised of the return earned on that security in its local currency, as well as a return from the performance of that foreign currency versus our domestic currency.) When a government pursues a long-term stance of expansionary fiscal policy, an investor should evaluate the implications of such a policy on expected risk and return (typically the yield should increase due to a higher risk premium). If investors perceive that the yield and/or currency return is sufficient, they will continue to purchase the bonds. However, continued increases in fiscal deficits are unsustainable and investors may refuse to fund the deficits—leading to currency depreciation.

Combining the Mundell-Fleming and portfolio balance approaches, we find that in the short term, with free capital flows, an expansionary fiscal policy leads to domestic currency appreciation (via high interest rates). In the long term, the government has to reverse course (through tighter fiscal policy) leading to depreciation of the domestic currency. If the government does not reverse course, it will have to monetize its debt (i.e., print money—monetary expansion), which would also lead to depreciation of the domestic currency.

LOS 5.l: Describe objectives of central bank or government intervention and capital controls and describe the effectiveness of intervention and capital controls.

A combination of "push" and "pull" factors determine the flow of capital into a country. Pull factors are favorable developments that make a country an attractive destination for foreign capital. These include relative price stability, a flexible exchange rate regime, improved fiscal position, privatization of state-owned enterprises etc. Push factors are largely driven by mobile international capital seeking high returns from a diversified portfolio.

As stated earlier, capital flows can lead to excessive appreciation of a currency. This can lead to several problems including loss of competitiveness of exports in the global markets, asset price bubbles, and excessive consumption fueled by credit creation. Excessive capital inflows to a country can also lead to a currency crisis when such capital is eventually withdrawn from the country. To reduce these problems, policymakers may intervene by imposing capital controls or by direct intervention in the foreign exchange market by the central bank.

Objectives

The objectives of capital controls or central bank intervention in FX markets are to:

- Ensure that the domestic currency does not appreciate excessively.
- Allow the pursuit of independent monetary policies without being hindered by their impact on currency values. For example, an emerging market central bank seeking to reduce inflation may pursue a restrictive monetary policy, increasing interest rates. However, these higher rates may attract large inflows of foreign capital, pushing up the value of the domestic currency.
- Reduce the aggregate volume of inflow of foreign capital.

Effectiveness

For developed market countries, the volume of trading in a country's currency is usually very large relative to the foreign exchange reserves of its central bank. Evidence has shown that for developed markets, central banks are relatively ineffective at intervening in the foreign exchange markets due to lack of sufficient resources. Evidence in the case of emerging markets is less clear: central banks of emerging market countries may be able to accumulate sufficient foreign exchange reserves (relative to trading volume) to affect the supply and demand of their currencies in the foreign exchange markets.

LOS 5.m: Describe warning signs of a currency crisis.

History has shown that market participants have failed to predict crises and typically are surprised by them. When market sentiment changes significantly, crises may occur *even for countries with sound economic fundamentals*.

The following conditions have been identified as warning signs in the period leading up to a currency crisis:

- Terms of trade (i.e., ratio of exports to imports) deteriorate.
- Fixed or partially-fixed exchange rates (versus floating exchange rates).
- Official foreign exchange reserves dramatically decline.
- Currency value that has risen above its historical mean.
- Inflation increases.
- Liberalized capital markets, that allow for the free flow of capital.
- Money supply relative to bank reserves increases.
- Banking crises (may also be coincident).

MODULE QUIZ 5.3

1. Vilasram Deshmukh is forecasting JPY/USD exchange rates based on balance of payments analysis. He notes that the United States is running large current account deficits relative to Japan. Based on this information, he concludes that the JPY/USD rate should decrease. His conclusion is *most likely* supported by the:
   A. flow mechanism of the current account influences.
   B. portfolio composition mechanism of the current account influences.
   C. capital account influences.
2. Stephen Hall is forecasting USD/GBP exchange rates. He consults forecasts of the money supply for the United States and U.K. made by his firm's chief economist, and he notes the following statement from a report published by the chief economist: "The U.S. money supply is expected to grow at a much faster pace than the U.K. or European money supplies."

   Hall makes the following statement: "Under the pure monetary approach model, an increase in the future growth rate of the money supply would lead to an immediate depreciation in the currency's value." Hall's statement is *most likely*:

   A. correct.
   B. incorrect, as the future growth rate in the money supply would not immediately affect currency values under the pure monetary approach model.
   C. incorrect, as the future growth rate in money supply would actually increase the currency value under the pure monetary approach.
3. Chintan Rajyaguru works for a currency dealer in London. He is evaluating the implications of changes in fiscal and monetary policies occurring in Zambola, an emerging market country with low capital mobility. He concludes that Zambola's central bank is pursuing a restrictive

monetary policy to curb inflation. Additionally, the Zambolan government has been reducing budget deficits to comply with new IMF lending terms. According to the Mundell-Fleming model, the change in monetary and fiscal policy is *most likely* to cause the Zambolan currency to:

A. appreciate.
B. depreciate.
C. remain unchanged.

Use the following information to answer Questions 4 through 9.

Agnetha Poulsen works as an analyst in the foreign exchange overlay strategies department for CFN, a large asset management firm serving institutional clients. She is concerned about the excessive unhedged currency exposure taken on by the overlay strategies department. She makes an appointment with Alvilda Kristensen, director of risk management, to discuss this matter. Prior to the meeting, Poulsen collects information on foreign currency quotes and on interest rates as shown in Figure 1 and Figure 2.

Figure 1: Current Spot and Forward Exchange Rate Quotes

Quotes	USD/CHF	USD/EUR
Spot	0.9817/0.9821	1.2235/1.2238
30-day forward	–7.6/–6.9	–7.21/–6.80
60-day forward	–15.3/–13.3	–14.56/–13.76
90-day forward	–24.3/–23.05	–23.84/–22.77

Figure 2: Selected Interest Rates

Interest Rates	USD	EUR	CHF
30-day rate	0.20%	0.91%	1.13%
60-day rate	0.21%	0.93%	1.15%
90-day rate	0.26%	1.04%	1.25%

Poulsen also reviews the current open forward contracts. As an example, she reviews two contracts. Contract FX2001 is a 90-day forward contract initiated 60 days ago. The contract calls for purchase of CHF 200 million at an all-in rate of USD 0.9832. Contract FX2051 is a 90-day contract initiated 30 days ago to purchase 100 million EUR at an all-in rate of 1.2242.

During her meeting with Kristensen, Poulsen expresses concern about traders establishing FX carry trades in several emerging market currencies. Kristensen assures Poulsen that CFN has adequate monitoring mechanisms. She continues that these carry trades have been generating significant positive returns for the clients and Poulsen should not worry about it. Poulsen counters by stating that carry trade returns distributions are characterized by negative kurtosis and excess skewness.

Poulsen reviews her notes and decides to prepare a report on currency crises. She compiles a list of indicators of an impending currency crisis based on empirical analysis.

Poulsen then turns her attention to the firm's investments in Zambola, an emerging market. She realizes that currently the currency overlay strategy department has no trades involving the free-floating Zambolan currency, the Zu. Poulsen is concerned about significant long exposure of the portfolio in Zu. Zambola is enjoying large capital inflows drawn by Zambola's attractive yields. Her analysis indicates that Zambola has been running large current account deficits. A trend analysis on Zu indicates a steep upward trend continuing above its PPP value.

1. The 30-day forward spread on USD/CHF is *closest* to:
   A. 0.0005.
   B. 0.0007.
   C. 0.7000.
2. The current mark-to-market value of the forward contract FX2001 in USD is *closest* to:
   A. –USD460,000.
   B. –USD451,924.
   C. –USD357,940.
3. The current mark-to-market value of the forward contract FX2051 in USD is *closest* to:
   A. –USD215,900.
   B. –USD107,900.
   C. –USD216,000.
4. Poulsen's description of the carry trade return distribution is *best* described as:
   A. correct.
   B. incorrect about skewness only.
   C. incorrect about both skewness and kurtosis.
5. Which of the following indicators of impending currency crises should Poulsen exclude from her report?
   A. Terms of trade improve.
   B. Increase in money supply relative to bank reserves.
   C. Increase in inflation.
6. If Zambolan government wanted to reduce the inflow of foreign capital, it should:
   A. pursue expansionary monetary policies.
   B. pursue policies consistent with currency appreciation.
   C. reduce inflation by increasing interest rates.

KEY CONCEPTS

LOS 5.a

bid–ask spread (for base currency) = ask quote − bid quote

Dealer spreads depend on spreads in the interbank market, the transaction size, and the dealer-client relationship. Interbank spreads depend on the currencies involved, time of day, and volatility in the currency pair. Forward spreads increase with maturities.

LOS 5.b

To calculate the profits from triangular arbitrage, start in the home currency and go around the triangle by exchanging the home currency for the first foreign currency, then exchanging the first foreign currency for the second foreign currency, and then exchanging the second foreign currency back into the home currency. If we end up with more money than what we had when we started, we've earned an arbitrage profit. The bid–ask spread forces us to buy a currency at a higher rate going one way than we can sell it for going the other way.

LOS 5.c

A spot exchange rate is for immediate delivery, while a forward exchange rate is for future delivery.

premium (discount) for base currency = forward price − spot price

LOS 5.d

The mark-to-market value of a forward contract reflects the profit that would be realized by closing out the position at current market prices, which is equivalent to offsetting the contract with an equal and opposite forward position:

$$V_t = \frac{(FP_t - FP)\,(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]}$$

where:

V_t = value of the forward contract at time t (to the party buying the base currency), ($t < T$) denominated in price currency

FP_t = forward price (to sell base currency) at time t in the market for a new contract maturing at time T

FP = forward price specified in the contract at inception (to buy the base currency)

days = number of days remaining to maturity of the forward contract ($T - t$)

R = interest rate of price currency

LOS 5.e

Covered interest arbitrage:

$$F = \frac{\left[1 + R_A\left(\frac{\text{days}}{360}\right)\right]}{\left[1 + R_B\left(\frac{\text{days}}{360}\right)\right]} S_0$$

Uncovered interest rate parity:

$$E(\%\Delta S)_{(A/B)} = R_A - R_B$$

International Fisher relation:

$$R_{\text{nominal A}} - R_{\text{nominal B}} = E(\text{inflation}_A) - E(\text{inflation}_B)$$

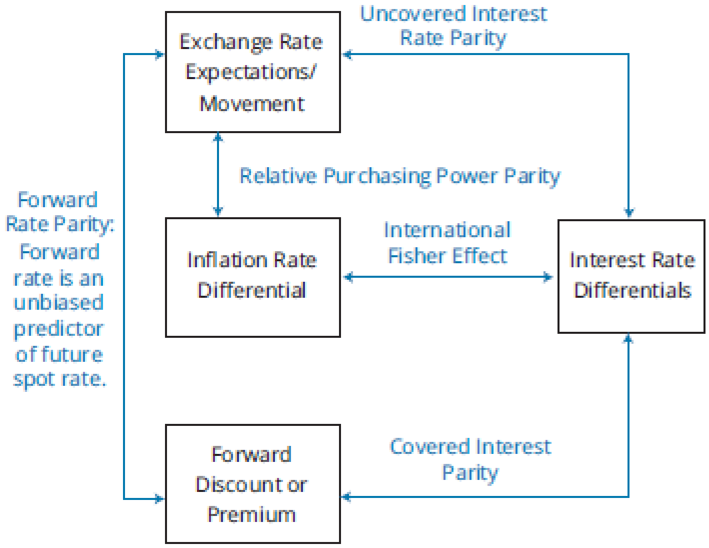
Relative PPP:

$$\%\Delta S_{(A/B)} = \text{inflation}_A - \text{inflation}_B$$

Forward rate parity:

$$F = E(S_T)$$

LOS 5.f



LOS 5.g, 5.h

Future spot rates can be forecasted using PPP or by uncovered interest rate parity. However, neither relationship is bound by arbitrage, nor do these relationships necessarily work in the short term. Forward exchange rates, on the other hand, can be estimated using covered interest parity, and this relationship is bound by arbitrage. If uncovered interest parity holds, then we say that the forward rate parity holds, i.e., the forward rate is an unbiased estimate of the future spot rate.

LOS 5.i

The FX carry trade seeks to profit from the failure of uncovered interest rate parity to work in the short run. In an FX carry trade, the investor invests in a high-yielding currency while borrowing in a low-yielding currency. If the higher yielding currency does not depreciate by the interest rate differential, the investor makes a profit. Carry trade has exposure to crash risk.

profit on carry trade = interest differential − change in the spot rate of the investment currency

LOS 5.j

BOP influence on exchange rate can be analyzed based on current account influence and capital account influence. Current account influences include flow mechanism, portfolio composition mechanism, and debt sustainability mechanism. Capital account inflows (outflows) are one of the major causes of increases (decreases) in exchange rates.

LOS 5.k

The Mundell-Fleming model of exchange rate determination evaluates the impact of monetary and fiscal policies on interest rates and consequently on exchange rates.

Under monetary models, we assume that output is fixed and, hence, monetary policies primarily affect inflation, which in turn affects exchange rates.

The portfolio balance (asset market) model evaluates the long-term implications of sustained fiscal policy (deficit or surplus) on currency values.

Monetary and Fiscal Policy and Exchange Rates

Monetary Policy/Fiscal Policy	Capital Mobility	
	High	Low
Expansionary/Expansionary	Uncertain	Depreciation
Expansionary/Restrictive	Depreciation	Uncertain
Restrictive/Expansionary	Appreciation	Uncertain
Restrictive/Restrictive	Uncertain	Appreciation

Under the pure monetary approach, PPP holds at any point in time.

Under the Dornbusch overshooting model, a restrictive (expansionary) monetary leads to an appreciation (depreciation) of domestic currency in the short term, and then slow depreciation (appreciation) towards the long-term PPP value.

Combining the Mundell-Fleming and portfolio balance approaches, we find that in the short term, an expansionary (restrictive) fiscal policy leads to domestic currency appreciation (depreciation). In the long term, the impact on currency values is opposite.

LOS 5.l

Capital controls and central bank intervention aim to reduce excessive capital inflows, which could lead to speculative bubbles. The success of central bank intervention depends on the size of official FX reserves at the disposal of the central bank relative to the average trading volume in the country's currency. For developed markets, the central bank resources on a relative basis are too insignificant to be effective at managing exchange rates. However, some emerging market countries with large FX reserves relative to trading volume have been somewhat effective.

LOS 5.m

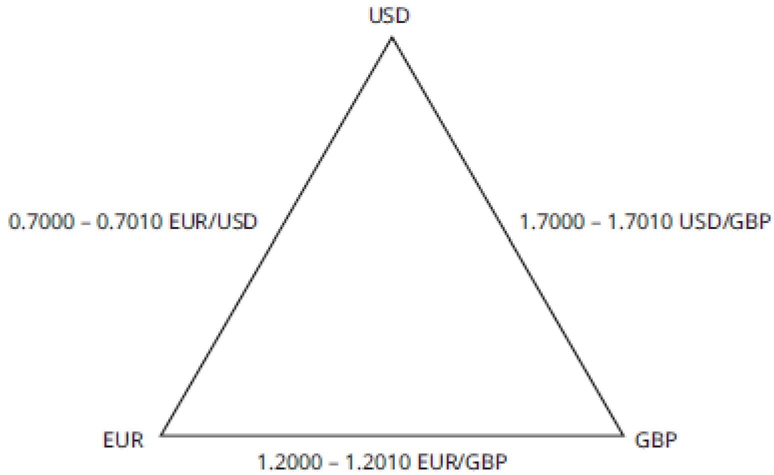
Warning signs of currency crises include: deterioration in terms of trade, a dramatic decline in official foreign exchange reserves, an exchange rate substantially higher than its mean-reverting level, increases in the inflation rate, a fixed- or partially-fixed exchange rate, an increase in money supply relative to bank reserves, and banking crises.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 5.1

1. **C** Dealer spreads are lower for smaller orders as compared to larger orders. Dealer spreads are larger when spreads in the interbank market are higher. An increase in spot rate volatility will increase spreads in the interbank market. (LOS 5.a)

2. **C** Here is what the triangle looks like with the bid–ask quotes filled in:



If we start with 1 million USD and move clockwise around the triangle (USD to GBP to EUR to USD), we first convert 1 million USD into GBP at the ask:

$$\frac{\text{1 million USD}}{\text{1.7010 USD/GBP}} = \text{587,889 GBP}$$

Then we sell the GBP for EUR at the bid:

$$\text{587,889 GBP} \times \left(\frac{\text{1.2000 EUR}}{\text{GBP}}\right) = \text{705,467 EUR}$$

Finally, we purchase USD at the ask in euros:

$$\frac{\text{705,467 EUR}}{0.7010} = \text{1,006,372 USD}$$

Arbitrage profits are 1,006,372 USD – 1,000,000 USD = 6,372 USD. (LOS 5.b)

Module Quiz 5.2

1. **A** Because of a lower interest rate, the USD (base currency) will appreciate by 2% to \$1.012 × 1.02 = C\$1.0322. (LOS 5.e)

2. **C** Combining all parity relationships indicates that the expected return on risk-free securities should be the same in all countries and exchange rate risk is really just inflation risk. There are four practical implications from this framework:
   1. The real, risk-free return will be the same in all countries.
   2. Investing in countries with high nominal interest rates will not generate excess returns because the high nominal interest rates will be accompanied by local currency depreciation.
   3. All investors will earn the same expected return in their own currency on any investment denominated in a foreign currency.
   4. Exchange rate risk is simply inflation risk, so investors interested in real returns will not face exchange rate risk.

   (LOS 5.f)

3. **A** Covered interest parity is forced by arbitrage, which is not the case for uncovered interest rate parity. If the forward rate is equal to the expected future spot rate, we say that the forward rate is an unbiased predictor of the future spot rate: $F = E(S_1)$. In this special case, given that covered interest parity holds, uncovered interest parity

would also hold (and vice versa). In other words, if uncovered interest rate parity (and covered interest parity) holds, the forward rate is unbiased predictor of future spot rate (i.e., forward rate parity holds). (LOS 5.e)

4. **A** According to the international Fisher relation:

$r = \text{real } r + E(I)$

From European data:

$4\% = \text{real } r + 2\%$

$\text{real } r = 2\%$

For United States:

$r = 2\% + 1\%$

$r = 3\%$

(LOS 5.e)

5. **A** Since inflation in Europe is higher than the inflation in the U.S. by 1%, the Euro is expected to depreciate by 1% annually against the dollar.

The current spot rate is \$(1/0.74) per Euro or \$1.3513/€.

expected exchange rate in 1 year = 1.3513(0.99) = \$1.3378/€

(LOS 5.e)

6. **C** Using covered interest parity, the forward rate in one year (in \$ per €) can be calculated as follows:

Spot rate = €0.74 per \$ = \$(1 _ 0.74) per €

$$F = S_0 \times \left(\frac{1 + r_\$}{1 + r_€}\right) = \left(\frac{1}{0.74}\right) \times \left(\frac{1.035}{1.04}\right) = \$1.3449 \text{ per } €$$

(LOS 5.e)

7. **B** Franklin is correct with respect to both of his statements: the rand should depreciate relative to the franc and the euro should depreciate relative to the dollar.

The relative form of purchasing power parity predicts that countries with higher expected inflation will experience a depreciation of their currencies. South Africa's expected inflation rate (5%) is higher than the expected inflation rate in Switzerland (3%). The expected inflation rate in Europe (2%) is higher than the expected inflation rate in the United States (1%). According to purchasing power parity, the rand should depreciate relative to the franc, and the euro should depreciate relative to the U.S. dollar.

Uncovered interest parity makes the same predictions with regard to relative interest rates: countries with higher nominal interest rates can be expected to experience currency depreciation. The South African interest rate (7%) is higher than the Swiss rate (5%), so uncovered interest rate parity predicts that the rand will depreciate with respect to the franc. The interest rate in Europe (4%) is higher than the interest rate in the United States (3%), so the euro should depreciate relative to the U.S. dollar. (LOS 5.e)

8. **C** According to the international Fisher relation, the real interest rate is equal to the nominal interest rate minus the expected inflation rate. The real interest rate in each of the four countries is 2%. (LOS 5.e)

9. **B** The 1-year expected spot rate should be equal to the current 1-year forward rate if uncovered interest rate parity holds. One of the assumptions of uncovered interest rate parity is that investors are risk neutral. Real interest rate parity states that real interest rates are equal across countries. Uncovered interest rate parity also would hold if both (1) relative (not absolute) PPP holds and (2) the international Fisher relationship holds. (LOS 5.e)

Module Quiz 5.3

1. **A** The flow mechanism of current account influences supports the view that current account deficits lead to depreciation of currency. In this example, the reduction in the JPY/USD rate implies depreciation of the USD. Under capital account influences, current account deficits imply capital account inflows and, hence, would lead to an appreciation of USD. The portfolio composition mechanism of current account influences supports the flow mechanism if investors rebalance a portion of their portfolio out of USD assets due to gradual buildup of USD assets over time in their portfolios. The question does not provide information to support this reallocation. (Module 5.3, LOS 5.j)

2. **B** Under the pure monetary approach, growth in the money supply leads to depreciation in currency. However, the future growth rate in money supply affects the trajectory of FX rates but not the current exchange rate. (Module 5.3, LOS 5.k)

3. **A** Under the Mundell-Fleming framework, low capital mobility and restrictive monetary and fiscal policy leads to better trade balance and appreciation of the country's currency. (Module 5.3, LOS 5.k)

4. **A** (0.9821 – 0.00069) – (0.9817 – 0.00076) = 0.00047

   (Module 5.1, LOS 5.c)

5. **B** The contract calls for purchase of 200 million CHF in 30 days. To compute the mark-to-market value, we would have to use the quote on 30-day forward contract to sell CHF. Given USD/CHF quote structure, we should use the bid price (going up the quote).

all-in bid price for 30-day USD/CHF forward contract =
0.9817 – 7.6 / 10,000 = 0.98094

$$V_t = \frac{(FP_t - FP)(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]}$$

$FP_t = 0.98094$ (computed above)
$FP = 0.9832$ (given)
R = 30-day USD interest rate (USD is the price currency)
= 0.20%

$$V_t = \frac{(0.98094 - 0.9832)(200{,}000{,}000)}{\left[1 + 0.002\left(\frac{30}{360}\right)\right]} = \frac{-452{,}000}{1.000166} = -451{,}924 \text{ USD}$$

(Module 5.2, LOS 5.d)

6. **A** The contract calls for purchase of 100 million EUR in 60 days. To compute the mark-to-market value, we would have to use the quote on 60-day forward contract to sell EUR. Given USD/EUR quote structure, we should use the bid price (going up the quote).

   all-in bid price for 60-day USD/EUR forward contract = 1.2235 – 14.56 / 10,000 = 1.22204

$$V_t = \frac{(FP_t - FP)(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]}$$

$FP_t = 1.22204$ (computed above)
$FP = 1.2242$ (given)
R = 60-day USD interest rate (USD is the price currency)
= 0.21%

$$V_t = \frac{(1.22204 - 1.2242)(100{,}000{,}000)}{\left[1 + 0.0021\left(\frac{60}{360}\right)\right]} = \frac{-216{,}000}{1.00035} = -215{,}924 \text{ USD}$$

(Module 5.2, LOS 5.d)

7. **C** Poulsen incorrectly described both the skewness as well as the kurtosis of carry trade returns. Carry trade return distributions generally have *negative skewness* and *excess kurtosis*. (Module 5.3, LOS 5.i)

8. **A** Deterioration (and not improvement) in terms of trade is an indicator of currency crisis. (Module 5.3, LOS 5.m)

9. **A** The Zu is overvalued per PPP, and Zambola is running a current account deficit. A depreciation of Zu would bring it closer to its long-run fair value. An increase in interest rates would lead to appreciation of Zu. Expansionary monetary policy would reduce interest rates and make Zambolan yields less attractive to foreign investors. (Module 5.3, LOS 5.j)

READING 6

ECONOMIC GROWTH

EXAM FOCUS

Forecasts of economic growth rates have important implications for investment decisions. Understand the preconditions of growth, how the growth rate can be increased, and what drives economic growth. Be able to compare and contrast competing theories of growth. Finally, be able to use growth accounting equations to forecast the potential growth rate of an economy.

MODULE 6.1: GROWTH FACTORS AND PRODUCTION FUNCTION

Video covering this content is available online.

LOS 6.a: Compare factors favoring and limiting economic growth in developed and developing economies.

Economists measure the economic output of a country by gross domestic product (GDP). A country's standard of living, however, is best measured by GDP per capita. Of particular concern to investors is not just the level of economic output but the growth rate of output.

Historically, there have been large variations in both GDP growth rates and per capita GDP across countries. Research has identified several factors that influence both the growth of GDP and the level of GDP.

Preconditions for Growth

1. **Savings and investment** are positively correlated with economic development. For countries to grow, private and public sector investment must provide a sufficient level of capital per worker. If a country has insufficient domestic savings, it must attract foreign investment in order to grow.
2. **Financial markets and intermediaries** augment economic growth by efficiently allocating resources in several ways. First, financial markets determine which potential users of capital offer the best returns on a risk-adjusted basis. Second, financial instruments are created by intermediaries that provide investors with liquidity and opportunities for risk reduction. Finally, by pooling small amounts of savings from investors, intermediaries can finance projects on larger scales than would otherwise be possible.

Some caution is in order, however. Financial sector intermediation may lead to declining credit standards and/or increases in leverage, increasing risk but not economic growth.

3. The **political stability, rule of law, and property rights** environment of a country also influence economic growth. Countries that have not developed a system of property rights for both physical and intellectual property will have difficulty attracting capital. Similarly, economic uncertainty caused by wars, corruption, and other disruptions poses unacceptable risk to many investors, reducing potential economic growth.
4. **Investment in human capital**, the investment in skills and well-being of workers, is thought to be complementary to growth in physical capital. Consequently, countries that invest in education and health care systems tend to have higher growth rates. Developed countries benefit the most from post-secondary education spending, which has been shown to foster innovation. Less-developed countries benefit the most from spending on primary and secondary education, which enables the workforce to *apply* the technology developed elsewhere.
5. **Tax and regulatory systems** need to be favorable for economies to develop. All else equal, the lower the tax and regulatory burdens, the higher the rate of economic growth. Lower levels of regulation foster entrepreneurial activity (startups), which have been shown to be positively related to the overall level of productivity.
6. **Free trade and unrestricted capital flows** are also positively related to economic growth. Free trade promotes growth by providing competition for domestic firms, thus increasing overall efficiency and reducing costs. Additionally, free trade opens up new markets for domestic producers. Unrestricted capital flows mitigate the problem of insufficient domestic savings as foreign capital can increase a country's capital, allowing for greater growth. Foreign capital can be invested directly in assets such as property, physical plant, and equipment (foreign direct investment), or invested indirectly in financial assets such as stocks and bonds.

LOS 6.b: Describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate of the economy.

Equity prices are positively related to earnings growth. Economy-wide, aggregate corporate earnings can grow if GDP grows or if the share of corporate earnings in GDP grows. However, the share of corporate profits in the GDP cannot increase indefinitely, because labor will be unwilling to work for a lower and lower proportion of the GDP. Therefore, the **potential GDP** of a country—the upper limit of *real* growth for an economy—is the long-run limit of earnings growth, and is an important factor in predicting returns on aggregate equity markets.

The Grinold-Kroner (2002) model provides the appropriate framework to understand this:

E(R) = dividend yield (DY) + expected capital gains yield (CGY)

We can further decompose the capital gains yield into EPS growth and growth in the P/E ratio:

expected capital gains yield = EPS growth (Δ EPS) + expected repricing (ΔP/E)

EPS growth can be decomposed into real EPS growth, inflation, and change in the number of shares outstanding via stock buybacks (i.e., the dilution effect).

$$\text{EPS growth} = \text{real EPS growth } (\Delta EPS_R) + \text{inflation } (\pi) - \text{change in shares outstanding } (\Delta S)$$

The full model then is:

$$E(R) = DY + \Delta EPS_R + \pi - \Delta S + \Delta P/E$$

Dividend yield tends to be stable over time. The repricing term (change in P/E) does fluctuate with the business cycle: when GDP growth is high, market multiples rise as perception of risk declines. The repricing term, however, cannot continue to grow. Therefore, in the long-run, the key factor driving equity returns is ΔEPS_R which is itself subject to an upper limit of potential GDP growth.

The dilution effect (ΔS) sometimes plays a role in determining equity returns, but its impact differs by country depending on its level of development and the sophistication of its financial markets. Recognize that on an aggregate basis, the dilution effect is comprised of net stock buybacks (nbb) as well as issuance by privately held small and medium entrepreneurial companies (it would actually be more appropriate to call *nbb* "net issuance," where net issuance = new issuance – buybacks). We call the role of these small and medium entrepreneurial companies the **relative dynamism (rd)** of the economy.

Therefore, $\Delta S = nbb + rd$

Relative dynamism captures the difference between the overall economic growth of the country and the earnings growth of listed companies.

LOS 6.c: Explain why potential GDP and its growth rate matter for equity and fixed income investors.

As indicated earlier, all else equal, higher GDP growth will be associated with higher equity returns. However, equity returns are significantly affected by dilution (due to share buybacks, mergers, privatizations of public companies, etc.). Economic growth can also be attributed to privately held companies. Potential GDP also has implications for real interest rates. Positive growth in potential GDP indicates that future income will rise relative to current income. When consumers expect their incomes to rise, they increase current consumption and save less for future consumption (i.e., they are less likely to worry about funding their future consumption). To encourage consumers to delay consumption (i.e., to encourage savings), investments would have to offer a higher real rate of return. Therefore, higher potential GDP growth implies higher real interest rates and higher real asset returns in general.

In the short term, the relationship between actual GDP and potential GDP may provide insight to both equity and fixed-income investors as to the state of the economy. For example, since actual GDP in excess of potential GDP results in rising prices, the gap between the two can be used as a forecast of inflationary pressures—useful to all investors but of particular concern to fixed-income investors. Furthermore, central banks are likely to adopt monetary policies consistent with the gap between potential output and actual output. When actual GDP growth rate is higher (lower) than potential GDP growth rate, concerns about inflation increase (decrease) and the central bank is more likely to follow a restrictive (expansionary) monetary policy.

In addition to predicting monetary policy, the relationship between actual and potential GDP can also be useful in analyzing fiscal policies. It is more likely for a government to run a fiscal deficit when actual GDP growth rate is lower than its potential growth rate.

Finally, because of the credit risk assumed by fixed-income investors, growth in GDP may be used to gauge credit risk of both corporate and government debt. A higher potential GDP growth rate reduces expected credit risk and generally increases the credit quality of all debt issues.

LOS 6.d: Contrast capital deepening investment and technological progress and explain how each affects economic growth and labor productivity.

Factor Inputs and Economic Growth

Economies are complex systems of many economic inputs. To simplify analysis, we examine a 2-factor (labor and capital) aggregate production function in which output (Y) is a function of labor (L) and capital (K), given a level of technology (T).

To examine the effect of capital investment on **economic growth** and **labor productivity**, consider a **Cobb-Douglas production function**, which takes the form:

$$Y = TK^{\alpha}L^{(1-\alpha)}$$

where:

α and $(1-\alpha)$ = the share of output allocated to capital (K) and labor (L), respectively [α and $(1-\alpha)$ are also referred to as capital's and labor's share of **total factor cost**, where $\alpha < 1$]

T = a scale factor that represents the **technological progress** of the economy, often referred to as *total factor productivity (TFP)*

The Cobb-Douglas function essentially states that output (GDP) is a function of labor and capital inputs and their productivity. It exhibits **constant returns to scale**; increasing both inputs by a fixed percentage leads to the same percentage increase in output.

Dividing both sides by L in the Cobb-Douglas production function, we can obtain the output per worker (labor productivity).

$$\text{output per worker} = Y/L = T(K/L)^{\alpha}$$

Labor productivity is similar to GDP per capita, a standard of living measure. The previous equation has important implications about the effect of capital investment on the standard of living. Assuming the number of workers and α remain constant, increases in output can be gained by increasing capital per worker (**capital deepening**) or by improving technology (increasing TFP).

However, since α is less than one, additional capital has a diminishing effect on productivity: the lower the value of α, the lower the benefit of capital deepening. Developed markets typically have a high capital to labor ratio and a lower α compared to developing markets, and therefore developed markets stand to gain less in increased productivity from capital deepening.

PROFESSOR'S NOTE

We need to distinguish between marginal product of capital and marginal productivity of capital. Marginal product of capital is the additional output for one additional unit of capital. Marginal productivity of capital is the increase in output per worker for one additional unit of capital per labor (i.e., increasing capital while keeping labor constant).

In steady state (i.e., equilibrium), the marginal product of capital ($MPK = \alpha Y/K$) and marginal cost of capital (i.e., the *rental price of capital*, r) are equal; hence:

$$\alpha Y/K = r$$

or

$$\alpha = rK/Y$$

PROFESSOR'S NOTE

In the previous equation, *r* is rate of return and *K* is amount of capital. *rK* measures the amount of return to providers of capital. The ratio of *rK* to output (Y) measures the amount of output that is allocated to providers of capital. This is precisely our definition of α.

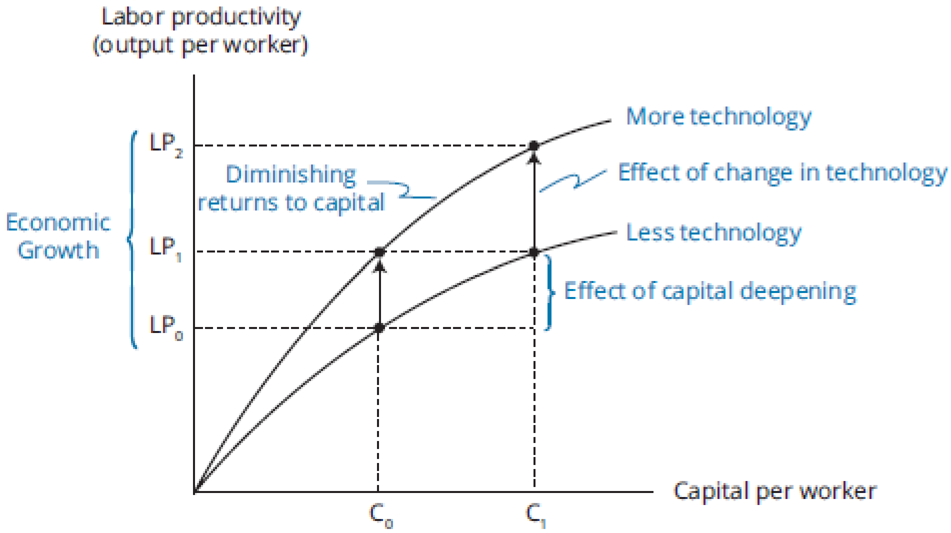
The productivity curves in Figure 6.1 show the effect of increasing capital per worker on output per worker. Capital deepening is a movement *along* the productivity curve. The curvature of the relationship derives from the diminishing marginal productivity of capital. Economies will increase investment in capital as long as MPK > r. At the level of K/L for which MPK = r, capital deepening stops and labor productivity becomes stagnant.

However, as technological progress occurs, both capital and labor can produce a higher level of output. An investment in capital leading to technological progress enhances the productivity of existing labor and capital. Technological progress, therefore, can lead to continued increases in output despite diminishing marginal productivity of capital. Technological progress *shifts* the productivity curve upward and will lead to increased productivity at all levels of capital per worker.

labor productivity growth rate

= growth due to technological change + growth due to capital deepening

Figure 6.1: Productivity Curves



As stated earlier, for developed countries, the capital per worker ratio is relatively high (e.g., level C_1 in Figure 6.1), so those countries gain little from capital deepening and must rely on technological progress for growth in productivity. In contrast, developing nations often have low capital per worker ratios (e.g., C_0 in Figure 6.1), so capital deepening can lead to at least a short-term increase in productivity.

MODULE 6.2: GROWTH ACCOUNTING AND INFLUENCING FACTORS

Video covering this content is available online.

LOS 6.e: Demonstrate forecasting potential GDP based on growth accounting relations.

Growth Accounting Relations

Using the Cobb-Douglas production function, the growth in potential GDP can be expressed using the **growth accounting relation** as:

$$\Delta Y/Y = \Delta T/T + \alpha \times (\Delta K/K) + (1-\alpha) \times (\Delta L/L)$$

where:

Y = output
T = technology
K = capital
L = labor
α = elasticity of output with respect to capital = share of income paid to capital
$(1 - \alpha)$ = elasticity of output with respect to labor = share of income paid to labor

or:

growth rate in potential GDP = long-term growth rate of technology
+ α (long-term growth rate of capital)
+ $(1 - \alpha)$ (long-term growth rate of labor)

In practice, levels of capital and labor are forecasted from their long-term trends, and the shares of capital and labor determined from national income accounts. The change in total factor productivity (technology) is not directly observable. Therefore, it must be estimated as a residual: the ex-post (realized) change in output minus the output implied by ex-post changes in labor and capital.

The growth accounting equation is also useful in determining the comparative effects of increasing different inputs. If labor growth accounts for the majority of economic growth, for example, analysts should be concerned with a country's ability to continue to increase its labor force. The relation can also be used to estimate potential output, as illustrated in the following example.

EXAMPLE: Estimating potential GDP growth rate

Azikland is an emerging market economy where labor cost accounts for 60% of total factor cost. The long-term trend of labor growth of 1.5% is expected to continue. Capital investment has been growing at 3%. The country has benefited greatly from borrowing the technology of more developed countries; total factor productivity is expected to increase by 2% annually. Compute the potential GDP growth rate for Azikland.

Answer:

Using the growth accounting equation:

growth rate in potential GDP = 2% + (0.4)(3%) + (0.6)(1.5%) = 4.1%

Another approach to forecasting potential GDP growth is the *labor productivity growth accounting equation*, which focuses on changes in labor as follows:

$$\text{growth rate in potential GDP} = \text{long-term growth rate of labor force} + \text{long-term growth rate in labor productivity}$$

The long-term growth rate in labor productivity reflects both capital deepening and technological progress.

LOS 6.f: Explain how natural resources affect economic growth and evaluate the argument that limited availability of natural resources constrains economic growth.

Natural resources include both renewable resources, such as timber, and non-renewable resources, such as oil and gas. The role of natural resources in economic growth is complex. In some instances, countries with abundant natural resources (e.g., Brazil) have grown rapidly. Yet other countries (e.g., some of the resource-rich countries of Africa) have not. Conversely, some resource-poor countries have managed impressive growth.

One reason that limited natural resources do not necessarily constrain economic growth is that *access* to natural resources does not require *ownership* of resources. Resource-poor countries may be able to access resources via trade. Japan, for example, has managed impressive growth and high per capita GDP despite having limited ownership of natural resources.

Other theories contend that ownership of natural resources may actually inhibit growth, because the economic energy of a country rich in natural resources may be focused on recovering those resources rather than developing other industries. Furthermore, countries that own valuable resources can find their currency appreciating as the demand for those resources increases. The so-called "Dutch disease" refers to a situation where global demand for a country's natural resources drives up the country's currency values, making all exports more expensive and rendering other domestic industries uncompetitive in the global markets.

LOS 6.g: Explain how demographics, immigration, and labor force participation affect the rate and sustainability of economic growth.

As stated previously, an increase in the quantity of labor will increase output, but not per capita output. Quantity of labor is defined as the size of the labor force multiplied by average hours worked. **Labor force** is defined as the number of working age (ages 16–64) people available to work, both employed and unemployed.

Labor Supply Factors

1. **Demographics.** A country's demographics strongly influence its potential economic growth. As a country's population ages and individuals live beyond working age, the labor force declines. Conversely, countries with younger populations have higher *potential* growth. Furthermore, fertility rates drive population growth and thereby affect potential future economic output. Countries with low or declining fertility rates will likely face growth challenges from labor force declines.
2. **Labor force participation.** Labor force participation is defined as the proportion of working age population in the labor force.

   $$\text{labor force participation} = \frac{\text{labor force}}{\text{working age population}}$$

   Labor force participation can increase as more women enter the workforce.
3. **Immigration.** Immigration poses a potential solution to a declining labor force. Countries with low population growth or adverse demographic shifts (older population) may find their growth constrained. Since developed countries tend to have lower fertility rates than less developed countries, immigration represents a potential source of continued economic growth in developed countries.
4. **Average hours worked.** For most countries, the general trend in average hours worked is downward. Possible explanations include legislation limiting the number of hours worked, the "wealth effect" which induces individuals to take more leisure time, high tax rates on labor income, and an increase in part-time and temporary workers.

EXAMPLE: Impact of demographics on economic growth

Data for Cangoria, a country in Asia, is shown here. Based upon this data, comment on the likely impact of Cangoria's demographic changes on its economic growth. Assume average world population growth rate is 1.2% per year.

	Population	Labor Force Participation	Median Age of Population
2013	23,400,400	60.4%	39.2
2023	28,040,300	70.3%	38.1

Answer:

Cangoria's population grew at an average annual compound growth rate of approximately 1.8% per year over those 10 years. Combined with the increase in labor force participation, labor supply growth should be above average in the future for Cangoria if those trends continue. The young median age of the population also indicates an expected increase in the labor pool in the future.

Changes in per capita GDP are difficult to predict. Output is expected to be higher due an increasing labor pool, but the larger population may mean there is no impact on per capita GDP.

LOS 6.h: Explain how investment in physical capital, human capital, and technological development affects economic growth.

Human capital. Human capital is knowledge and skills individuals possess. Unlike quantitative labor metrics, such as hours worked, human capital is a qualitative measure of the labor force. Increasing human capital through education or work experience increases productivity and economic growth. Furthermore, human capital may have external spillover effects as knowledgeable workers innovate. Innovations are then used by society in general creating greater efficiencies economy wide.

Physical capital. Physical capital is generally separated into infrastructure, computers, and telecommunications capital (ICT) and non-ICT capital (i.e., machinery, transportation, and non-residential construction). Empirical research has found a strong positive correlation between investment in physical capital and GDP growth rates.

This result may seem inconsistent given our previous discussion about capital deepening and diminishing marginal returns to capital. Several explanations exist to explain why capital increases may still result in economic growth. First, many countries (e.g., developing economies) have relatively low capital to labor ratios, so increases in capital may still have significant impact on economic growth. Second, capital investment can take different forms. Some capital investment actually influences technological progress, thereby increasing TFP and economic growth. For example, acceleration of spending in the IT sector has created what are termed *network externalities*. Investment in IT networks may have multiplicative effects on productivity since IT network investment actually becomes more valuable as more people are connected to the network.

Technological development. Investment in technology includes investment in both physical and human capital. Technological innovation can manifest itself in processes, knowledge, information, machinery, and software, among other things. Researchers have examined proxies for investment in technology such as research and development (R&D) spending or number of patents issued. Developed countries tend to spend the most on R&D since they rely on technological progress for growth given their high existing capital stock and slower population growth. In contrast, less developed countries often copy the technological innovations of developed countries and thus invest less in R&D as a percentage of GDP.

Ultimately, technological development should lead to increases in productivity as measured by GDP per worker. Developed countries tend to have very high levels of productivity by this measure while less developed countries tend to have greater potential for growth in productivity.

Public infrastructure. Investments in public infrastructure such as the construction of public roads, bridges, and municipal facilities, provide additional benefits to *private* investment. For example, an investment in distribution facilities by a private company would do little good without an interstate highway grid. The highway system, therefore, enhances total productivity for the economy by complementing the private investment and increasing total factor productivity.

MODULE QUIZ 6.1, 6.2

Use the following information to answer Questions 1 through 6.

Jay Smith, an analyst for Mako Capital, is evaluating investment prospects in Minikaz, an emerging market economy. Minikaz has experienced moderate growth in the past four years, after decades of stagnation. Smith is evaluating changes in government policies that would foster a higher level of growth. Figure 1 shows the summary of his findings.

Figure 1: Proposed Changes in Minikaz Government Policies

1. Consumer protection will be at the forefront of government's agenda.
2. The government will lower the entry barriers for foreign financial institutions to operate as intermediaries in Minikaz capital markets.
3. The government will expand public domain legislation to acquire private property for public works projects.

Smith reviews a report published by the Minikaz commerce department. The report indicates that the long-term real growth rate of Minikaz GDP is 2.5%, corporate profits as a percentage of GDP increased by 2% last year, and the P/E ratio increased from 17 to 19 over the last two years. Separately, Smith also reviews World Bank reports indicating that Minikaz's potential GDP growth is 4% and that it has been experiencing actual GDP growth of approximately 2.5%. Finally, Smith reviews Minikaz's national income accounts and finds that Minikaz is experiencing both technological progress and making increased capital expenditures.

Separately, Smith evaluates the performance of Kinimaz, a neighboring republic. Kinimaz has had labor growth of 2% over the last several years and capital growth of 3%. Labor's share of total output is estimated to be 60%. Over the same period, Kinimaz's real GDP has grown by 3.7%. Comparing the two countries, Smith notes that Kinimaz has substantially higher amounts of natural resource endowments. He concludes that Minikaz's relatively lower GDP growth is due to lack of natural resources.

1. Which of the following actions by Minikaz's government is most likely to increase Minikaz's economic growth rate?
   A. Increasing protection for consumers through regulations.
   B. Allowing foreign financial institutions to enter the market.

C. Expanding public domain legislation.

2. Based on the commerce department report, what would be the *most likely* forecast for the long-term aggregate stock market appreciation?
   A. 2.5%.
   B. 4.5%.
   C. 11.5%.
3. Based on World Bank report, which of the following conclusions is *most likely* regarding Minikaz?
   A. Inflation is 1.5%.
   B. Minikaz's government is likely to follow a restrictive fiscal policy.
   C. Minikaz's central bank is not likely to be worried about inflation.
4. Using the Cobb-Douglas production function and the concepts of capital deepening and total factor productivity, which of the following outcomes is *most likely*?
   A. Minikaz will experience an increase in sustainable growth of per capita output due to the increased capital expenditures.
   B. There will be no short-term increase in per capita output.
   C. There will be both short-term and long-term increases in Minikaz's GDP growth rate.
5. Using the Cobb-Douglas relation, total factor productivity growth for Kinimaz is *closest* to:
   A. 0.5%.
   B. 1.3%.
   C. 1.7%.
6. Smith's conclusion about Minikaz's relatively lower GDP growth is *most likely*:
   A. correct.
   B. correct because in some cases, natural resources may inhibit economic growth.
   C. incorrect because access to natural resources is more important than ownership.
7. Data for the labor market of countries X and Y over the past year appears next:

Country	Unemployment Rate	% Population < Age 15	Avg. Hours Worked/Week	Immigration Growth
X	16%	16%	37	3.5%
Y	3%	10%	36.5	3.0%

   Both countries are expected to have moderate economic expansions over the next several years. Which of the following statements is *most accurate* regarding labor input of the countries in the next several years?

   A. Country X will have greater opportunities to increase labor input.
   B. Country Y will have greater opportunities to increase labor input.
   C. Neither Country X nor Country Y will be able to increase labor input.
8. Which of the following would *least likely* have externality effects on output growth for an economy?
   A. Human capital investment.
   B. ICT investment.
   C. Non-ICT investment.

MODULE 6.3: GROWTH AND CONVERGENCE THEORIES

Video covering this content is available online.

LOS 6.i: Compare classical growth theory, neoclassical growth theory, and

endogenous growth theory.

Theories of economic growth are largely separated into three models with differing views on the steady state growth potential of an economy.

Classical Growth Theory

Based on Malthusian economics, classical growth theory posits that, in the long-term, population growth increases whenever there are increases in per capita income above subsistence level due to an increase in capital or technological progress. Subsistence level is the minimum income needed to maintain life. Classical growth theory contends that growth in real GDP per capita is not permanent, because when real GDP per capita rises above the subsistence level, a population explosion occurs. Population growth leads to diminishing marginal returns to labor, which reduces productivity and drives GDP per capita back to the subsistence level. This mechanism would prevent long-term growth in per capita income. Classical growth theory is not supported by empirical evidence.

Neoclassical Growth Theory

Neoclassical growth theory's primary focus is on estimating the economy's long-term **steady state growth rate** (sustainable growth rate or equilibrium growth rate). The economy is at equilibrium when the output-to-capital ratio is constant. When the output-to-capital ratio is constant, the capital-to-labor ratio and output per capita also grow at the equilibrium growth rate, g*. Under neoclassical theory, population growth is independent of economic growth.

PROFESSOR'S NOTE
Steady state growth rate for the purpose of neoclassical growth theory does not assume a constant level of technology and hence differs from the definition of steady state discussed earlier.

Based on the Cobb-Douglas function discussed earlier, neoclassical growth theory states that:

- Sustainable growth of output per capita (or output per worker)(g*) is equal to the growth rate in technology (θ) divided by labor's share of GDP (1 – α).

$$g^* = \frac{\theta}{(1-\alpha)}$$

- Sustainable growth rate of output (G*) is equal to the sustainable growth rate of output per capita, plus the growth of labor (ΔL).

$$G^* = \frac{\theta}{(1-\alpha)} + \Delta L$$

PROFESSOR'S NOTE
In the equations for sustainable growth (per capita or total), growth rate is not affected by capital (K). Hence, we say that capital deepening is occurring but it does not affect growth rate once steady state is achieved.

EXAMPLE: Estimating steady state growth rate

An analyst is forecasting steady state growth rates for Country X and Country Y and has collected the following estimates:

Country	TFP Growth Rate	Labor Force Growth Rate	Labor Cost as a Proportion of Total Factor Cost
X	2.0%	1.2%	0.60
Y	1.0%	2.6%	0.52

Calculate and comment on sustainable growth rates for the two countries.

Answer:

Sustainable growth rates:

Country X = (2.0% / 0.60) + 1.2% = 4.53%

Country Y = (1.0% / 0.52) + 2.6% = 4.52%

Thus, the sustainable growth rates for the two countries are comparable. Country X's sustainable growth rate is primarily driven by higher growth rate in TFP. Country Y's sustainable growth rate is mostly driven by a higher population growth rate.

Under neoclassical theory:

- Capital deepening affects the *level* of output but not the *growth rate* in the long run. Capital deepening may temporarily increase the growth rate, but the growth rate will revert back to the sustainable level if there is no technological progress.
- An economy's growth rate will move towards its steady state regardless of the *initial* capital to labor ratio or level of technology.
- In the steady state, the growth rate in productivity (i.e., output per worker) is a function only of the growth rate of technology (θ) and labor's share of total output ($1 - \alpha$).
- In the steady state, marginal product of capital (MPK) = $\alpha Y/K$ is constant, but marginal productivity is diminishing.
- An increase in savings will only temporarily raise economic growth. However, countries with higher savings rates will enjoy higher capital to labor ratio and higher productivity.
- Developing countries (with a lower level of capital per worker) will be impacted less by diminishing marginal productivity of capital, and hence have higher growth rates as compared to developed countries; there will be eventual convergence of growth rates.

Endogenous Growth Theory

In contrast to the neoclassical model, **endogenous growth theory** contends that technological growth emerges as a *result* of investment in both physical and human capital (hence the name *endogenous* which means coming from within). Technological progress enhances productivity of both labor and capital. Unlike the neoclassical model, there is no steady state growth rate, so that increased investment can permanently increase the rate of growth.

The driving force behind the endogenous growth theory result is the assumption that certain investments increase TFP (i.e., lead to technological progress) from a societal standpoint. Increasing R&D investments, for example, results in benefits that are also external to the firm making the R&D investments. Those benefits raise the level of growth for the entire economy.

The endogenous growth model theorizes that returns to *capital* are constant. The key implication of constant returns to capital is the effect of an increase in savings: unlike the neoclassical model, the endogenous growth model implies that an increase in savings will permanently increase the growth rate.

The difference between neoclassical and endogenous growth theory relates to total factor productivity. Neoclassical theory assumes that capital investment will expand as technology improves (i.e., growth comes from increases in TFP not related to the investment in capital within the model). Endogenous growth theory, on the other hand, assumes that capital investment (R&D expenditures) may actually improve total factor productivity.

LOS 6.j: Explain and evaluate convergence hypotheses.

Empirical evidence indicates that there are large differences between productivity (output per capita) of different countries, with less developed countries experiencing much lower output per capita than their developed counterparts. The economic question is whether productivity, and hence, living standards tend to converge over time. Will less developed countries experience productivity growth to match the productivity of developed nations?

The **absolute convergence** hypothesis states that less-developed countries will converge to the level of per capita output of more-developed countries. The neoclassical model assumes that every country has access to the same technology. This leads to countries having the same *growth rates* but not the same per capita income and as such, the neoclassical model does not imply absolute convergence. The neoclassical model supports the **conditional convergence** hypothesis, which states that the convergence in living standards will only occur for countries with the same savings rates, population growth rates, and production functions. Under the conditional convergence hypothesis, the growth rate will be higher for less developed countries until they catch up and achieve a similar standard of living. Under the neoclassical model, once a developing country's standard of living converges with that of developed countries, the growth rate will then stabilize to the same steady state growth rate as that of developed countries.

An additional hypothesis is **club convergence**. Under this hypothesis, countries may be part of a "club" (i.e., countries with similar institutional features such as savings rates, financial markets, property rights, health and educational services, etc.). Under club convergence, poorer countries that are part of the club will grow rapidly to catch up with their richer peers. Countries can "join" the club by making appropriate institutional changes. Those countries that are not part of the club may never achieve the higher standard of living.

Empirical evidence shows that developing economies often (but not always) reach the standard of living of more developed ones. Over the past half century, about two-thirds of economies with a lower standard of living than the United States grew at a faster pace than the United States. Though they have not converged to standard of living of the United States, their more rapid growth provides at least some support for the convergence hypothesis. The club convergence theory may explain why some countries that have not implemented appropriate economic or political reforms still lag behind.

LOS 6.k: Describe the economic rationale for governments to provide incentives to private investment in technology and knowledge.

Firms accept projects when they provide an expected return greater than their risk-adjusted cost of capital. Under endogenous growth theory, private sector investments in R&D and knowledge capital benefit the society overall. For example, a new technology may initially benefit the firm that developed it but may also boost the country's overall productivity. The effects of "social returns" or externalities are captured in the endogenous growth theory model, which concludes that economies may not reach a steady state growth but may permanently increase growth by expenditures that provide both benefits to the company (private benefits) and benefits to society (externalities).

When the external benefits to the economy (the social returns) of investing in R&D are not considered, many possible R&D projects do not have expected returns (private benefits) high enough to compensate firms for the inherent riskiness of R&D investments. From an aggregate, economy-wide viewpoint, the resultant level of R&D investment will be sub-optimal or too low. Government incentives that effectively subsidize R&D investments can theoretically increase private spending on R&D investments to its optimal level.

LOS 6.l: Describe the expected impact of removing trade barriers on capital investment and profits, employment and wages, and growth in the economies involved.

None of the growth theories that we have discussed account for potential trade and capital flows between countries. Removing trade barriers and allowing for free flow of capital is likely to have the following benefits for countries:

- Increased investment from foreign savings.
- Allows focus on industries where the country has a comparative advantage.
- Increased markets for domestic products, resulting in economies of scale.
- Increased sharing of technology and higher total factor productivity growth.
- Increased competition leading to failure of inefficient firms and reallocation of their assets to more efficient uses.

The neoclassical model's predictions in an open economy (i.e., an economy without any barriers to trade or capital flow) focus on the convergence. Since developing economies have not reached the point of significant diminishing returns on capital, they can attract capital through foreign investment and experience productivity growth as a result. Eventually, these economies will develop; their growth will slow and will converge to the steady state growth rate of developed economies.

The endogenous growth model also predicts greater growth with free trade and high mobility of capital since open markets foster increased innovation. As foreign competition increases, more efficient and innovative firms will survive. Those firms permanently increase the growth rate of the international economy by providing benefits beyond those simply captured by the firm. Economies of scale also increase output as firms serve larger markets and become more efficient.

In terms of convergence, removing barriers on capital and trade flows may speed the convergence of standard of living of less developed countries to that of developed countries. Research has shown that as long as countries follow outward-oriented policies of integrating their industries with the world economy and increasing exports, their standard of living tends to converge to that of more developed countries. Countries following inward-oriented policies and protecting domestic industries, can expect slower GDP growth and convergence may not occur.

MODULE QUIZ 6.3

1. Country X has output elasticity of capital of 0.6 and population growth of 2%. If total factor productivity growth is 1%, what is the sustainable growth rate in output according to neoclassical theory?
   A. 2.0%.
   B. 2.7%.
   C. 4.5%.
2. Which of the following is the *most accurate* description of club convergence?
   A. Less developed countries will converge to living standards of other less developed countries.
   B. More developed countries may see their standard of living drop due to competition from less developed countries.
   C. Some less developed countries may converge to developed country living standards while others may not.
3. A chief economist argues that government policy should include an additional tax break for research and development expenses. The economist *most likely* agrees with:
   A. endogenous growth theory.
   B. neoclassical theory.
   C. classical theory.

Use the following information to answer Questions 4 through 5.

Jignesh Sangani, an economist with a large asset management firm, makes the following statements about removal of barriers to trade and capital flows:

Statement 1: Removal of barriers is likely to lead to permanently higher global economic growth under the neoclassical theory.

Statement 2: Removal of barriers is likely to lead to permanently higher economic growth for developing countries only under the endogenous growth theory.

4. Sangani's Statement 1 is *most likely*:
   A. correct.
   B. incorrect due to economic growth being permanent.
   C. incorrect due to economic growth being global.
5. Sangani's Statement 2 is *most likely*:
   A. correct.

B. incorrect due to economic growth being permanent.
C. incorrect due to economic growth being limited to developing countries only.

6. Which of the following is *least likely* to be associated with the law of diminishing returns?
   A. Investment in labor.
   B. Investment in knowledge capital.
   C. Investment in physical capital.

KEY CONCEPTS

LOS 6.a

Significant differences in growth rates exist between economies. The following factors are positively related to growth rate:

- Sufficient level of savings and investment.
- Development of financial markets and financial intermediaries.
- Political stability, sound laws, and property rights.
- Investment in education and health care systems.
- Lower taxes and regulatory burdens.
- Free trade and unrestricted capital flows.

LOS 6.b

In the long-run, the rate of aggregate stock market appreciation is limited to the sustainable growth rate of the economy.

LOS 6.c

Potential GDP represents the maximum output of an economy without putting upward pressure on prices. Higher potential GDP growth increases the potential for stock returns but also increases the credit quality of all fixed-income investments, all else equal.

In the short term, the difference between potential GDP and actual GDP may be useful for predicting fiscal and monetary policy. If actual GDP is less than potential GDP, inflation is unlikely and the government may follow an expansionary monetary/fiscal policy.

LOS 6.d

Capital deepening is an increase in the capital stock and the capital to labor ratio. Due to diminishing marginal productivity of capital, capital deepening will lead to only limited increases in output and labor productivity if the capital to labor ratio is already high.

Technological progress enhances the productivity of both labor and capital but not the relative productivity of either. The long-term growth rate can be increased by technological progress (also called total factor productivity) since output and labor efficiency are increased at all levels of capital to labor ratios. TFP is a scale factor. It however does not alter the productivity function (alpha value).

LOS 6.e

$$\begin{aligned}\text{growth rate in potential GDP} &= \text{long-term growth rate of technology} \\ &+ \alpha\,(\text{long-term growth rate in capital}) \\ &+ (1-\alpha)\,(\text{long-term growth rate in labor})\end{aligned}$$

or

$$\begin{aligned}\text{growth rate in potential GDP} &= \text{long-term growth rate of labor force} \\ &+ \text{long-term growth rate in labor productivity}\end{aligned}$$

LOS 6.f

Natural resources are essential to economic growth. Empirical evidence has shown, however, that *ownership* of natural resources is not necessary for growth. As long as nations can acquire natural resources through trade, they can experience substantial growth. In some cases, ownership of natural resources may even inhibit growth since countries with abundant natural resources may not develop other industries.

LOS 6.g

Quantity of labor is a function of population growth, workforce participation, immigration, and average hours worked. All else equal, countries with higher population growth, higher workforce participation, younger working-age populations, higher average hours worked, and higher net immigration can grow faster due to higher labor input.

LOS 6.h

The economic growth rate of a country is positively correlated with investments in both physical and human capital. Furthermore, technological development (as evidenced by spending on R&D) is critical for economic growth. This is especially true for developed countries that already have large capital stock and a slower population growth rate.

LOS 6.i

Classical growth theory states that growth in real GDP per capita is temporary—when the GDP per capita rises above the subsistence level, a population explosion occurs, and GDP per capita is driven back to the subsistence level.

Neoclassical growth theory states that the sustainable growth rate of GDP is a function of population growth, labor's share of income, and the rate of technological advancement. Growth gains from other means such as increased savings are only temporary.

Endogenous growth theory includes the impact of technological progress within the model. Under endogenous growth theory, investment in capital can have constant returns, unlike neoclassical theory that assumes diminishing returns to capital. This assumption allows for a permanent increase in growth rate attributable to an increase in savings rate. Research and development expenditures are often cited as examples of capital investment that increase technological progress.

LOS 6.j

Absolute convergence states that the per capita growth *rates* (not growth *level)* will converge (i.e., be the same across all countries). The conditional convergence hypothesis assumes that convergence in living standards (i.e., *level* of per capita output) will only occur for countries with the same savings rate, population growth, and production functions.

The club convergence hypothesis contends that living standards in some less developed countries may converge to living standards of developed standards if they are in the same "club." A club comprises countries with similar institutional structures (such as property rights and political stability). Countries outside of the club (without the appropriate institutional structures) will not see their living standards converge.

LOS 6.k

Under the endogenous growth theory, investments in R&D, though risky, often enhance the productivity of the entire economy. Since the private investor only reaps part of the benefit of those investments, it is likely that private sector investments in R&D will be less than what would be optimal for the economy. Government subsidies can make these investments more attractive to private businesses.

LOS 6.l

Economies grow faster in an environment of no trade barriers and free capital flows. Higher growth rates are possible because foreign investment can provide capital to less developed countries (neoclassical theory). The larger markets and greater opportunity to take advantage of innovation will also increase the growth rate in open economies (endogenous growth theory).

Finally, convergence of living standards is likely to be quicker in an open economy.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 6.1, 6.2

1. **B** Financial intermediary development helps foster economic growth by allowing more efficient allocation of capital and risk. (Module 6.1, LOS 6.a)

2. **A** Long-term growth in the stock market is a function of GDP growth. The other factors —profits as a percentage of GDP and P/E ratios—will have a long-term growth rate of approximately zero and will not impact a forecast of long-term growth in the stock market. (Module 6.1, LOS 6.a)

3. **C** Potential GDP can be interpreted as the highest growth that can be obtained without pressure on prices. Since actual GDP is lower than potential, there is little risk of inflation. (Module 6.1, LOS 6.b)

4. **C** Since Minikaz is a developing country, it is likely to have a low capital base. With a low capital base, increased capital expenditures will still have an impact on output per worker. Technological progress always has a positive impact on output per worker. (Module 6.1, LOS 6.d)

5. **B** Use the growth accounting relations and solving for growth in TFP.

$$3.7\% = \Delta TFP + 0.4(3\%) + 0.6(2\%)$$
$$\Delta TFP = 1.3\%$$

(Module 6.2, LOS 6.e)

6. **C** Empirical evidence has shown that for economic growth, access to natural resources is more important than ownership. Natural resources may inhibit growth if countries that own them do not develop other industries. However, that is not the conclusion Smith reaches. (Module 6.2, LOS 6.f)

7. **A** Country X will have the greater opportunity due to the younger workforce, potential labor input from unemployed workers, and immigration. (Module 6.2, LOS 6.g)

8. **C** Both human capital and ICT investment tend to have societal benefits. This spillover effect enhances overall growth rate. (Module 6.2, LOS 6.h)

Module Quiz 6.3

1. **C** Using the equation from neoclassical theory, 1% / (1 − 0.6) + 2% = 4.5%. (LOS 6.i)

2. **C** The notion of the club is that some nations are not in the club and will not converge. (LOS 6.j)

3. **A** Endogenous growth theory includes the concept that R&D may have external benefits, and, therefore, should be subsidized by the government. (LOS 6.i)

4. **B** Under the neoclassical growth theory, the benefit of open markets is temporary. (LOS 6.i)

5. **C** Under the endogenous growth theory, open markets lead to higher rate of growth permanently for all markets. (LOS 6.i)

6. **B** Knowledge capital is a special type of public good that is not subject to the law of diminishing returns. Investment in labor and physical capital do exhibit diminishing returns, which are reflected in the shape of the productivity curve. (LOS 6.k)

Topic Quiz: Economics

You have now finished the Economics topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow three minutes per question.

FORMULAS

Quantitative Methods

Multiple Regression

Coefficient of Determination, R^2

$$R^2 = \frac{\text{total variation} - \text{unexplained variation}}{\text{total variation}} = \frac{SST - SSE}{SST}$$

$$= \frac{\text{explained variation}}{\text{total variation}} = \frac{RSS}{SST}$$

$$MSE = \frac{SSE}{n-k-1};\ MSR = \frac{RSS}{k};\ R^2 = \frac{RSS}{SST}$$

Adjusted R^2

$$R_a^2 = 1 - \left[\left(\frac{n-1}{n-k-1}\right) \times (1 - R^2)\right]$$

Akaike's information criterion (AIC):

$$AIC = n \times \ln\left(\frac{SSE}{n}\right) + 2(k+1)$$

Schwarz's Bayesian information criteria (BIC):

$$BIC = n \times \ln\left(\frac{SSE}{n}\right) + \ln(n) \times (k+1)$$

***F*-statistic to evaluate nested models:**

$$F = \frac{(SSE_R - SSE_U)/q}{(SSE_U)/(n-k-1)} \textit{ with q and (n – k – 1) degrees of freedom.}$$

***F*-test statistic to evaluate *overall* model fit:**

$$F = \frac{(SSE_R - SSE_U)/q}{(SSE_U)/(n-k-1)} = \frac{(SST_U - SSE_U)/k}{(SSE_U)/(n-k-1)} = \frac{(RSS_U)/k}{(SSE_U)/(n-k-1)}$$

Logistic regression (logit) models

$$\ln\left(\frac{p}{1-p}\right) = b_0 + b_1X_1 + b_2X_2 + \ldots + \varepsilon$$

$$\text{odds} = e^{\hat{y}}$$

$$P = \text{odds} / (1 + \text{odds}) = 1/(1 + e^{-\hat{y}})$$

Likelihood ratio (LR) test for logistic regressions:

LR = –2 (log likelihood restricted model – log likelihood unrestricted model)

Time-Series Analysis

AR model of order *p*, AR(*p*): $x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-2} + \ \ldots \ + b_p x_{t-p} + \varepsilon_t$

Mean reverting level of AR(1): $x_t = \dfrac{b_0}{(1 - b_1)}$

ARCH(1) model: $\hat{\varepsilon}_t^2 = a_0 + a_1 \hat{\varepsilon}_{t-1}^2 + \mu_t$

Big Data Projects

$$\text{normalized } X_i = \frac{X_i - X_{min}}{X_{max} - X_{min}}$$

$$\text{standardized } X_i = \frac{X_i - \mu}{\sigma}$$

accuracy = (TP + TN) / (TP + TN + FP + FN)

F1 score = (2 × P × R) / (P + R)

true positive rate (TPR) = TP / (TP + FN)

false positive rate (FPR) = FP / (FP + TN)

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} (\text{predicted}_i - \text{actual}_i)^2}{n}}$$

Economics

Where applicable, ALL notation assumes A/B currency quote convention.

bid–ask spread (for base currency) = ask quote – bid quote

cross rates with bid–ask spreads:

$$\left(\frac{A}{C}\right)_{bid} = \left(\frac{A}{B}\right)_{bid} \times \left(\frac{B}{C}\right)_{bid} \qquad \left(\frac{A}{C}\right)_{offer} = \left(\frac{A}{B}\right)_{offer} \times \left(\frac{B}{C}\right)_{offer}$$

forward premium = (forward price) – (spot price) = $F - S_0$

value of a forward currency contract prior to expiration:

$$V_t = \frac{(FP_t - FP)(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]}$$

covered interest rate parity:

$$F = \frac{\left[1 + R_A\left(\frac{\text{days}}{360}\right)\right]}{\left[1 + R_B\left(\frac{\text{days}}{360}\right)\right]} S_0$$

uncovered interest rate parity:

$$E(\%\Delta S)_{(A/B)} = R_A - R_B$$

Fisher relation:

$$R_{nominal} = R_{real} + E(\text{inflation})$$

international Fisher relation:

$$R_{nominal\ A} - R_{nominal\ B} = E(\text{inflation}_A) - E(\text{inflation}_B)$$

relative purchasing power parity:

$$\%\Delta S_{(A/B)} = \text{inflation}_{(A)} - \text{inflation}_{(B)}$$

where:

$\%\Delta S_{(A/B)}$ = change in spot price (A/B)

labor productivity:

$$\text{output per worker} = Y/L = T(K/L)^{\alpha}$$

growth accounting relation:

$$\begin{aligned}\text{growth rate in potential GDP} &= \text{long-term growth rate of technology} \\ &\quad + \alpha(\text{long-term growth rate of capital}) \\ &\quad + (1 - \alpha)(\text{long-term growth rate of labor})\end{aligned}$$

or

$$\begin{aligned}\text{growth rate in potential GDP} &= \text{long-term growth rate of labor force} \\ &\quad + \text{long-term growth rate in labor productivity}\end{aligned}$$

neoclassical growth theory:

sustainable growth of output per capita (g*) equals growth rate in technology (θ) divided by labor's share of GDP ($1 - \alpha$)

$$g^* = \frac{\theta}{(1 - \alpha)}$$

sustainable growth rate of output (G*) equals sustainable growth rate of output per capita plus growth of labor (ΔL)

$$G^* = \frac{\theta}{(1 - \alpha)} + \Delta L$$

APPENDIX A: STUDENT'S T-DISTRIBUTION

	Level of Significance for One-Tailed Test					
df	0.100	0.050	0.025	0.01	0.005	0.0005
	Level of Significance for Two-Tailed Test					
df	0.20	0.10	0.05	0.02	0.01	0.001
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.599
3	1.638	2.353	3.182	4.541	5.841	12.294
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.869
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.408
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.768
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.045	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.358	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

APPENDIX B: F-TABLE AT 5% (UPPER TAIL)

Degrees of freedom for the numerator along top row

Degrees of freedom for the denominator along side row

	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40
1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	6.01	2.94	2.90	2.86	2.83
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39

APPENDIX C: CHI-SQUARED TABLE

Values of χ^2 (Degrees of Freedom, Level of Significance)

Probability in Right Tail

Degrees of Freedom	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000157	0.000982	0.003932	0.0158	2.706	3.841	5.024	6.635	7.879
2	0.020100	0.050636	0.102586	0.2107	4.605	5.991	7.378	9.210	10.597
3	0.1148	0.2158	0.3518	0.5844	6.251	7.815	9.348	11.345	12.838
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.554	0.831	1.145	1.610	9.236	11.070	12.832	15.086	16.750
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	4.107	5.009	5.892	7.041	19.812	22.362	24.736	27.688	29.819
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.558
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	12.878	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.994
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.335
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
100	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.170

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